HMoE report

HMoE Funnel Routing & Precision: Results Report

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1. Headline result

1.0 Which comparison is which

Three different 'routed vs dense' comparisons are possible here and they answer different questions; conflating them is the easiest way to over-read this report.

(a) Full budget (both arms near the 10M cap). Best full-recipe routed run `e5-seed-2` reaches EM 0.526 at 9,607,384 parameters against the dense baseline's 0.500 at 9,965,120 — i.e. at equal budget the routed model is +0.026. Section 3 and findings 1-4 analyse this arm. The single best full-budget routed cell overall is the ablation `e4-ablate-no-proxy-warmup` at 0.527; it is reported separately because it removes a collapse mitigation rather than adding one. **(b) Reduced budget (~3.5M total), default hyperparameters.** Best sparse routed run `e8-sparse-d192` EM 0.411 at 3,677,355 vs the size-matched dense `e9-dense-matched-3p5m` EM 0.495 at 3,505,536: -0.083. This is the number that looks like 'routing costs 10 points', and on its own it is not trustworthy — see (c). **(c) Reduced budget, both arms tuned over an identical grid (E10).** Best tuned routed cell `e10-sparse-lr0.5-long-s2` (sparse shape, 7 cells) EM 0.450 vs best tuned dense cell `e10-dense-lr1-long` (7 cells) EM 0.653: -0.204. Both sides had the same number of grid cells at the same schedule, so this is the like-for-like arm and the only one that supports an architecture claim. Section 12 gives the full paired grid and separates capacity, optimisation and generalisation.

run	variant	params (total)	params active/ token	active / dense total	EM(tf)	EM(free)	VE(free)	PC(free)	PC(tf)	EM(soft-route)	free/ TF agree	H_final	guardrail hits	wall (s)	vs dense
e1-noise-s0.1-exp	funnel	9607384	9441368	0.947	0.522	0.522	0.522	0.778	0.809	0.521	1.000	0.316	0	261.1	+0.022
e1-noise-s0.1-linear	funnel	9607384	9441368	0.947	0.502	0.502	0.502	0.768	0.804	0.503	1.000	0.348	0	261.1	+0.002
e1-noise-s0.3-exp	funnel	9607384	9441368	0.947	0.525	0.525	0.525	0.777	0.808	0.526	1.000	0.346	0	261.9	+0.025
e1-noise-s0.3-linear	funnel	9607384	9441368	0.947	0.520	0.520	0.520	0.777	0.811	0.519	1.000	0.290	0	262.0	+0.020
e1-noise-s0.5-exp	funnel	9607384	9441368	0.947	0.522	0.522	0.523	0.784	0.813	0.524	1.000	0.299	0	267.1	+0.022
e1-noise-s0.5-linear	funnel	9607384	9441368	0.947	0.524	0.524	0.524	0.773	0.808	0.524	1.000	0.453	0	264.6	+0.024
e1-noise-s0.8-exp	funnel	9607384	9441368	0.947	0.524	0.524	0.524	0.780	0.812	0.522	1.000	0.291	0	265.7	+0.024
e1-noise-s0.8-linear	funnel	9607384	9441368	0.947	0.516	0.516	0.516	0.774	0.808	0.516	1.000	0.411	0	265.9	+0.016
e10-dense-lr0.25-s1	dense	3505536	-	-	0.440	0.440	0.440	0.706	0.758	-	1.000	-	0	128.4	-0.060
e10-dense-lr0.25-s2	dense	3505536	-	-	0.446	0.446	0.446	0.706	0.758	-	1.000	-	0	128.3	-0.054
e10-dense-lr0.5-long-s1	dense	3505536	-	-	0.524	0.524	0.524	0.777	0.805	-	1.000	-	0	311.9	+0.024
e10-dense-lr0.5-long-s2	dense	3505536	-	-	0.535	0.535	0.535	0.778	0.804	-	1.000	-	0	311.4	+0.035
e10-dense-lr0.5-s1	dense	3505536	-	-	0.472	0.472	0.472	0.738	0.777	-	1.000	-	0	136.8	-0.028
e10-dense-lr0.5-s2	dense	3505536	-	-	0.491	0.491	0.491	0.750	0.787	-	1.000	-	0	131.7	-0.009
e10-dense-lr1-long	dense	3505536	-	-	0.653	0.653	0.653	0.852	0.869	-	1.000	-	0	314.1	+0.153
e10-dense-lr1-long-s1	dense	3505536	-	-	0.602	0.602	0.602	0.828	0.848	-	1.000	-	0	309.1	+0.102
e10-dense-lr1-long-s2	dense	3505536	-	-	0.604	0.604	0.604	0.831	0.849	-	1.000	-	0	313.9	+0.104
e10-dense-lr1-s1	dense	3505536	-	-	0.542	0.542	0.542	0.787	0.815	-	1.000	-	0	129.2	+0.042
e10-dense-lr1-s2	dense	3505536	-	-	0.535	0.535	0.535	0.791	0.814	-	1.000	-	0	131.6	+0.035
e10-dense-lr1.5-long-s1	dense	3505536	-	-	0.000	0.000	0.000	0.044	0.034	-	1.000	-	0	284.2	-0.500
e10-dense-lr1.5-long-s2	dense	3505536	-	-	0.000	0.000	0.000	0.240	0.240	-	1.000	-	0	283.7	-0.500
e10-dense-lr1.5-s1	dense	3505536	-	-	0.000	0.000	0.000	0.301	0.301	-	1.000	-	0	121.8	-0.500
e10-dense-lr1.5-s2	dense	3505536	-	-	0.506	0.506	0.506	0.777	0.810	-	1.000	-	0	129.0	+0.006
e10-dense-lr2-s1	dense	3505536	-	-	0.000	0.000	0.000	0.050	0.050	-	1.000	-	0	115.8	-0.500
e10-dense-lr2-s2	dense	3505536	-	-	0.000	0.000	0.000	0.301	0.301	-	1.000	-	0	117.7	-0.500
e10-sparse-d128-lr0.25-s1	funnel	3514943	3014824	0.303	0.306	0.306	0.306	0.633	0.681	0.274	1.000	0.666	0	137.0	-0.194
e10-sparse-d128-lr0.25-s2	funnel	3514943	3014824	0.303	0.296	0.296	0.296	0.628	0.671	0.268	1.000	0.656	0	137.4	-0.204
e10-sparse-d128-lr0.5-long-s1	funnel	3514943	3014824	0.303	0.438	0.438	0.438	0.691	0.739	0.435	1.000	0.347	0	327.5	-0.062
e10-sparse-d128-lr0.5-long-s2	funnel	3514943	3014824	0.303	0.438	0.438	0.438	0.687	0.733	0.431	1.000	0.373	0	327.9	-0.062
e10-sparse-d128-lr0.5-s1	funnel	3514943	3014824	0.303	0.394	0.394	0.394	0.664	0.718	0.366	1.000	0.431	0	136.7	-0.106
e10-sparse-d128-lr0.5-s2	funnel	3514943	3014824	0.303	0.379	0.379	0.379	0.658	0.705	0.366	1.000	0.496	0	136.3	-0.121
e10-sparse-d128-lr1-long-s1	funnel	3514943	3014824	0.303	0.455	0.455	0.455	0.714	0.756	0.448	1.000	0.322	0	328.7	-0.045
e10-sparse-d128-lr1-long-s2	funnel	3514943	3014824	0.303	0.461	0.461	0.461	0.722	0.761	0.449	1.000	0.319	0	329.1	-0.039
e10-sparse-d128-lr1-s1	funnel	3514943	3014824	0.303	0.403	0.403	0.403	0.672	0.726	0.390	1.000	0.352	0	136.7	-0.097
e10-sparse-d128-lr1-s2	funnel	3514943	3014824	0.303	0.410	0.410	0.410	0.677	0.722	0.390	1.000	0.343	0	137.2	-0.090
e10-sparse-d128-lr1.5-long-s1	funnel	3514943	3014824	0.303	0.035	0.035	0.035	0.417	0.464	0.034	1.000	0.915	0	325.5	-0.465
e10-sparse-d128-lr1.5-long-s2	funnel	3514943	3014824	0.303	0.030	0.030	0.030	0.424	0.463	0.030	1.000	0.864	0	328.6	-0.470
e10-sparse-d128-lr1.5-s1	funnel	3514943	3014824	0.303	0.344	0.344	0.344	0.639	0.684	0.317	1.000	0.471	0	137.2	-0.156
e10-sparse-d128-lr1.5-s2	funnel	3514943	3014824	0.303	0.373	0.373	0.373	0.662	0.706	0.314	1.000	0.413	0	136.8	-0.127
e10-sparse-d128-lr2-s1	funnel	3514943	3014824	0.303	0.008	0.008	0.008	0.202	0.275	0.008	1.000	1.004	0	126.9	-0.492
e10-sparse-d128-lr2-s2	funnel	3514943	3014824	0.303	0.023	0.023	0.023	0.399	0.437	0.023	1.000	1.002	0	345.6	-0.477
e10-sparse-lr0.25-s1	funnel	3677355	3243848	0.326	0.359	0.359	0.359	0.651	0.697	0.351	1.000	0.588	0	137.6	-0.141
e10-sparse-lr0.25-s2	funnel	3677355	3243848	0.326	0.373	0.373	0.373	0.665	0.709	0.360	1.000	0.593	0	136.6	-0.127
e10-sparse-lr0.5-long-s1	funnel	3677355	3243848	0.326	0.449	0.449	0.449	0.701	0.746	0.444	1.000	0.398	0	334.5	-0.051
e10-sparse-lr0.5-long-s2	funnel	3677355	3243848	0.326	0.450	0.450	0.450	0.706	0.746	0.447	1.000	0.384	0	342.3	-0.050
e10-sparse-lr0.5-s1	funnel	3677355	3243848	0.326	0.406	0.406	0.406	0.670	0.718	0.392	1.000	0.369	0	137.7	-0.094
e10-sparse-lr0.5-s2	funnel	3677355	3243848	0.326	0.415	0.415	0.415	0.680	0.725	0.403	1.000	0.383	0	136.7	-0.085
e10-sparse-lr1-long	funnel	3677355	3243848	0.326	0.421	0.421	0.421	0.692	0.740	0.377	1.000	0.352	0	330.5	-0.079
e10-sparse-lr1-long-s1	funnel	3677355	3243848	0.326	0.025	0.025	0.025	0.438	0.490	0.042	1.000	0.417	1	329.9	-0.475

run	variant	params (total)	params active/ token	active / dense total	EM(tf)	EM(free)	VE(free)	PC(free)	PC(tf)	EM(soft-route)	free/ TF agree	H_final	guardrail hits	wall (s)	vs dense
e10-sparse-lr1-long-s2	funnel	3677355	3243848	0.326	0.447	0.447	0.447	0.714	0.764	0.407	1.000	0.378	0	330.2	-0.053
e10-sparse-lr1-s1	funnel	3677355	3243848	0.326	0.408	0.408	0.408	0.679	0.723	0.393	1.000	0.387	0	139.0	-0.092
e10-sparse-lr1-s2	funnel	3677355	3243848	0.326	0.408	0.408	0.408	0.686	0.732	0.386	1.000	0.400	0	137.8	-0.092
e10-sparse-lr1.5-long-s1	funnel	3677355	3243848	0.326	0.009	0.009	0.009	0.205	0.326	0.009	1.000	0.512	0	309.4	-0.491
e10-sparse-lr1.5-long-s2	funnel	3677355	3243848	0.326	0.000	0.000	0.000	0.309	0.325	0.000	1.000	0.472	0	301.8	-0.500
e10-sparse-lr1.5-s1	funnel	3677355	3243848	0.326	0.025	0.025	0.025	0.400	0.439	0.027	1.000	0.768	0	135.9	-0.475
e10-sparse-lr1.5-s2	funnel	3677355	3243848	0.326	0.024	0.024	0.024	0.399	0.444	0.025	1.000	0.653	0	137.4	-0.476
e10-sparse-lr2-s1	funnel	3677355	3243848	0.326	0.002	0.002	0.002	0.239	0.261	0.002	1.000	0.542	0	131.1	-0.498
e10-sparse-lr2-s2	funnel	3677355	3243848	0.326	0.021	0.021	0.021	0.394	0.433	0.019	1.000	0.706	0	142.9	-0.479
e2-dense-baseline	dense	9965120	9965120	1.000	0.500	0.500	0.500	0.759	0.794	-	1.000	-	0	232.4	+0.000
e3-oracle-capacity-4x	funnel_capacity	12068824	11410520	1.145	0.473	0.473	0.473	0.739	0.792	0.474	1.000	0.360	0	302.9	-0.027
e3-oracle-routing-trained	funnel	9607384	9441368	0.947	0.521	0.521	0.521	0.775	0.809	0.449	1.000	-	0	249.5	+0.021
e3-oracle-tied-experts	funnel_tied	9109336	8943320	0.897	0.507	0.507	0.507	0.769	0.806	0.509	1.000	0.583	0	267.0	+0.007
e4-ablate-flat-temp-B	funnel	9607384	9441368	0.947	0.508	0.508	0.508	0.767	0.801	0.510	1.000	0.413	0	266.2	+0.008
e4-ablate-gumbel	funnel	9607384	9441368	0.947	0.523	0.523	0.523	0.780	0.811	0.521	1.000	0.400	0	266.7	+0.023
e4-ablate-instant-hard	funnel	9607384	9441368	0.947	0.524	0.524	0.524	0.777	0.809	0.521	1.000	0.140	14	267.2	+0.024
e4-ablate-no-aux	funnel	9607384	9441368	0.947	0.517	0.517	0.517	0.775	0.809	0.518	1.000	0.188	2	266.1	+0.017
e4-ablate-no-proxy-warmup	funnel	9607384	9441368	0.947	0.527	0.527	0.527	0.779	0.811	0.526	1.000	0.191	7	266.4	+0.027
e4-ablate-no-router-freeze	funnel	9607384	9441368	0.947	0.509	0.509	0.509	0.768	0.800	0.508	1.000	0.349	0	266.6	+0.009
e5-seed-1	funnel	9607384	9441368	0.947	0.521	0.521	0.521	0.775	0.812	0.520	1.000	0.421	0	266.2	+0.021
e5-seed-2	funnel	9607384	9441368	0.947	0.526	0.526	0.526	0.784	0.815	0.528	1.000	0.442	0	266.0	+0.026
e6-scale-n6000	funnel	9607384	9441368	0.947	0.304	0.304	0.304	0.601	0.653	0.305	1.000	0.369	0	257.1	-0.196
e6-scale-n60000	funnel	9607384	9441368	0.947	0.593	0.593	0.593	0.826	0.857	0.589	1.000	0.402	0	270.6	+0.093
e7-iso-active	funnel	9981728	9774047	0.981	0.526	0.526	0.526	0.780	0.815	0.525	1.000	0.439	0	268.7	+0.026
e8-sparse-d128	funnel	3514943	3014824	0.303	0.397	0.397	0.397	0.669	0.719	0.368	1.000	0.450	0	136.1	-0.103
e8-sparse-d160	funnel	3727477	3227362	0.324	0.399	0.399	0.399	0.662	0.704	0.379	1.000	0.404	0	144.7	-0.101
e8-sparse-d192	funnel	3677355	3243848	0.326	0.411	0.411	0.411	0.669	0.720	0.399	1.000	0.347	0	138.6	-0.089
e8-sparse-d192-wide	funnel	5028705	4362273	0.438	0.431	0.431	0.431	0.695	0.738	0.425	1.000	0.403	0	159.1	-0.069
e9-dense-matched-3p5m	dense	3505536	3505536	0.352	0.495	0.495	0.495	0.750	0.792	-	1.000	-	0	130.4	-0.005

`EM(tf)` teacher-forced slot exact match; `EM(free)` free-running greedy decode (the headline number); `VE` value-exact (numeric equality); `PC` partial credit over answer slots only; `EM(soft-route)` evaluates the same weights with routing averaged over experts instead of argmax'd; `free/TF agree` is the fraction of examples where both evaluation paths give the same verdict (exposure-bias check, 1.0 = none). `params (total)` is what the <=10M budget caps; `params active/token` is what actually participates in producing one token under hard [3,2,2,1] routing, and `active / dense total` puts that against the dense baseline's per-token cost (equal to its total). Section 10 breaks both down per funnel stage.

Best full-budget routed run on the baseline corpus **e4-ablate-no-proxy-warmup** reaches free-running exact-match **0.527** against the dense baseline's **0.500** (+0.027) at a smaller parameter count (9607384 vs 9965120). Reduced-budget (E8/E9/E10) and data-scaling (E6) runs are excluded here: they answer different questions, and section 1.0 says which comparison supports which claim.

1.1 Findings (computed from the logged data)

1. **Routing vs dense.** Under the same <=10M budget the routed funnel `e5-seed-2` beats the dense baseline on free-running exact match: 0.526 vs 0.500 (+0.026). Spread across the swept routing configs: 0.502-0.526.
2. **Annealing schedule.** Averaged over matched noise scales, the exponential sigma decay is better: linear mean 0.516, exponential mean 0.523 (n=4 each). Best single cell: `e1-noise-s0.3-exp` at 0.525.
3. **Noise scale.** Best sigma0 = 0.3 (max EM 0.525); ordering by best-case EM: sigma=0.3:0.525, sigma=0.5:0.524, sigma=0.8:0.524, sigma=0.1:0.522.
4. **Entropy vs precision.** Final routing entropy spans 0.290 (`e1-noise-s0.3-linear`) to 0.453 (`e1-noise-s0.5-linear`) while EM spans 0.502-0.525; the two orderings do not agree, i.e. 'sharper routing' is not by itself a proxy for accuracy at this scale.
5. **Pure routing cost.** On `e1-noise-s0.1-exp` - trained without ever seeing the true paths - injecting ground-truth `[3,2,2]` paths at evaluation moves EM from 0.522 to 0.517 (-0.005) with the weights and tile capacity held fixed, and perfect routing does **worse** than the learned router, so the learned router is not leaving anything on the table: at this budget its choices are already as good as ground truth (the tiles carry the task, the partition does not).
6. **Routing oracle (trained).** `e3-oracle-routing-trained`, with true paths forced at every stage *during training*, reaches 0.521 vs the self-taught routed model's 0.524: knowing the correct partition from the start buys -0.003.
7. **Capacity vs routing.** The 4x-width capacity oracle reaches 0.473 at 12,068,824 params - *over* the 10M budget by design - against the in-budget routed model's 0.524; the tied-expert capacity floor gives 0.507. Extra width actually *hurt*: 0.052 worse at 1.3x the parameters. The larger tiles under-train on this corpus rather than helping, so extra width is not the lever here - see the data-scaling finding for what is.
8. **Data scaling (same parameter budget).** Corpus size is the only lever that moved accuracy by a large amount: 6,000 training rows -> EM 0.304, default corpus (6,000 test rows scored) -> 0.524, 60,000 -> 0.593 (parameter count identical at 9,607,384 in all three cells), i.e. +0.289 from data alone, against -0.052 from 4x the parameters (finding on capacity). The scaling triggers for a 10M->100M expansion (spec 7.4) are therefore *not* met: exhaust the corpus before spending parameters.
9. **Collapse mitigations.** Largest degradations relative to the headline config (0.524): `e4-ablate-flat-temp-B` 0.508 (-0.016); `e4-ablate-no-router-freeze` 0.509 (-0.015); `e4-ablate-no-aux` 0.517 (-0.007).
10. **Guardrails.** 4 of 78 runs tripped a guardrail (`e10-sparse-lr1-long-s1` x1, `e4-ablate-instant-hard` x14, `e4-ablate-no-aux` x2, `e4-ablate-no-proxy-warmup` x7). Logged reasons: 23 entropy-floor events, 1 utilisation-ceiling event; peak single-path utilisation over learned-routing runs was 62% (`e10-sparse-lr1.5-long-s2`, live stages only) - well below the 70% starvation ceiling, i.e. no expert burned out. Runs with ground-truth routing forced are excluded: they concentrate all traffic on one expert per stage by construction. 3 of 4 runs that tripped one were collapse-mitigation ablations, which is the expected direction; the remaining 1 (`e10-sparse-lr1-long-s1`) were full-recipe or tuning runs, so at least some configurations did need an intervention.
11. **Evaluation integrity.** Free-running and teacher-forced verdicts agree on 100.0-100.0% of examples across all runs, so the reported exact-match numbers are not inflated by teacher forcing.
12. **Cost.** Wall time 265s for the routed run vs 232s dense on the same steps/data (1.14x), single RTX 5060 Laptop GPU. **Sparse vs dense at matched small budget.** This is the only arm where both sides got an identical tuning grid (7 cells each at 12,000 steps): routed `e10-sparse-lr0.5-long-s2` 0.450 vs dense `e10-dense-lr1-long` 0.653, i.e. routing *costs* 20.4 EM points at this budget. The untuned version of this comparison looks smaller (+8 to +10 points) and is not the one to quote. Section 12 separates how much is capacity, optimisation and generalisation; the short answer there is that both arms can reach near-zero training loss at

this size, so the routed arm's deficit is credit assignment under this optimiser budget, not a capacity ceiling.

2. Fixed experimental setup

item	value
total parameter cap	10,000,000 (hard)
funnel parameters (measured from checkpoints)	3,514,943 / 3,677,355 / 3,727,477 / 5,028,705 / 9,607,384 / 9,981,728
dense baseline parameters (measured)	3,505,536 / 9,965,120
tied-expert oracle parameters (measured)	9,109,336
capacity-oracle parameters (measured; over budget by design)	12,068,824
routing parameters (measured, funnel runs)	39,224 / 47,736 / 56,248 / 442,264
runs exceeding the 10M cap	e3-oracle-capacity-4x (12,068,824) - capacity oracle only, which spec 5.1 asks for as an over-budget upper bound; no headline or comparison model exceeds the cap
tokenizer	fixed table, 119 tokens, no learned BPE
domains / tiers	BIN t1-3, HEX t1-3, CHEM t1-3 (9 generators)
funnel shape	4 -> 3 -> 2 -> 1 tiles; [3,2,2] cascade gating
val / test rows actually scored	3,000 val / 6,000 test rows scored per run
EM(free)	free-running greedy decode, no teacher forcing
EM(tf)	teacher-forced slot exact match over answer positions

2.1 Configuration log (every knob of every run)

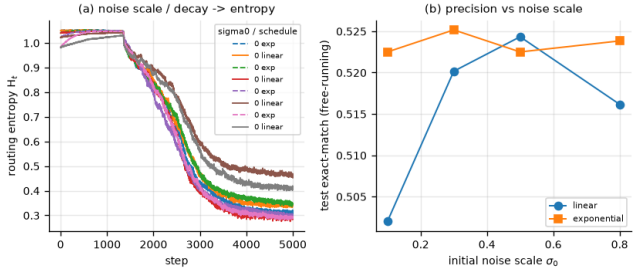
run	variant	steps	lr	warmup_frac	harden_frac	router_freeze_frac	sigma0	noise_decay	T_hi	T_lo	temp_in_B	gumbel	aux_alpha	aux_beta	proxy_weight	entropy_floor	util_ceiling	seed
e1-noise-s0.1-exp	funnel	5000	0.0006	0.12	0.55	0.35	0.1	exp	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e1-noise-s0.1-linear	funnel	5000	0.0006	0.12	0.55	0.35	0.1	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e1-noise-s0.3-exp	funnel	5000	0.0006	0.12	0.55	0.35	0.3	exp	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e1-noise-s0.3-linear	funnel	5000	0.0006	0.12	0.55	0.35	0.3	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e1-noise-s0.5-exp	funnel	5000	0.0006	0.12	0.55	0.35	0.5	exp	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e1-noise-s0.5-linear	funnel	5000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e1-noise-s0.8-exp	funnel	5000	0.0006	0.12	0.55	0.35	0.8	exp	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e1-noise-s0.8-linear	funnel	5000	0.0006	0.12	0.55	0.35	0.8	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e10-dense-lr0.25-s1	dense	5000	0.0003	0.12	0.55	0.35	0.0	const	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-dense-lr0.25-s2	dense	5000	0.0003	0.12	0.55	0.35	0.0	const	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-dense-lr0.5-long-s1	dense	12000	0.0006	0.12	0.55	0.35	0.0	const	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-dense-lr0.5-long-s2	dense	12000	0.0006	0.12	0.55	0.35	0.0	const	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-dense-lr0.5-s1	dense	5000	0.0006	0.12	0.55	0.35	0.0	const	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-dense-lr0.5-s2	dense	5000	0.0006	0.12	0.55	0.35	0.0	const	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-dense-lr1-long	dense	12000	0.0012	0.12	0.55	0.35	0.0	const	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e10-dense-lr1-long-s1	dense	12000	0.0012	0.12	0.55	0.35	0.0	const	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-dense-lr1-long-s2	dense	12000	0.0012	0.12	0.55	0.35	0.0	const	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-dense-lr1-s1	dense	5000	0.0012	0.12	0.55	0.35	0.0	const	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-dense-lr1-s2	dense	5000	0.0012	0.12	0.55	0.35	0.0	const	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-dense-lr1.5-long-s1	dense	12000	0.0018	0.12	0.55	0.35	0.0	const	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-dense-lr1.5-long-s2	dense	12000	0.0018	0.12	0.55	0.35	0.0	const	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-dense-lr1.5-s1	dense	5000	0.0018	0.12	0.55	0.35	0.0	const	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-dense-lr1.5-s2	dense	5000	0.0018	0.12	0.55	0.35	0.0	const	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-dense-lr2-s1	dense	5000	0.0024	0.12	0.55	0.35	0.0	const	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-dense-lr2-s2	dense	5000	0.0024	0.12	0.55	0.35	0.0	const	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-sparse-d128-lr0.25-s1	funnel	5000	0.0003	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-sparse-d128-lr0.25-s2	funnel	5000	0.0003	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-sparse-d128-lr0.5-long-s1	funnel	12000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-sparse-d128-lr0.5-long-s2	funnel	12000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2

run	variant	steps	lr	warmup_frac	harden_frac	router_freeze_frac	sigma0	noise_decay	T_hi	T_lo	temp_in_B	gumbel	aux_alpha	aux_beta	proxy_weight	entropy_floor	util_ceiling	seed
e10-sparse-d128-lr0.5-s1	funnel	5000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-sparse-d128-lr0.5-s2	funnel	5000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-sparse-d128-lr1-long-s1	funnel	12000	0.0012	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-sparse-d128-lr1-long-s2	funnel	12000	0.0012	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-sparse-d128-lr1-s1	funnel	5000	0.0012	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-sparse-d128-lr1-s2	funnel	5000	0.0012	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-sparse-d128-lr1.5-long-s1	funnel	12000	0.0018	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-sparse-d128-lr1.5-long-s2	funnel	12000	0.0018	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-sparse-d128-lr1.5-s1	funnel	5000	0.0018	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-sparse-d128-lr1.5-s2	funnel	5000	0.0018	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-sparse-d128-lr2-s1	funnel	5000	0.0024	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-sparse-d128-lr2-s2	funnel	5000	0.0024	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-sparse-lr0.25-s1	funnel	5000	0.0003	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-sparse-lr0.25-s2	funnel	5000	0.0003	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-sparse-lr0.5-long-s1	funnel	12000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-sparse-lr0.5-long-s2	funnel	12000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-sparse-lr0.5-s1	funnel	5000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-sparse-lr0.5-s2	funnel	5000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-sparse-lr1-long	funnel	12000	0.0012	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e10-sparse-lr1-long-s1	funnel	12000	0.0012	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-sparse-lr1-long-s2	funnel	12000	0.0012	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-sparse-lr1-s1	funnel	5000	0.0012	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-sparse-lr1-s2	funnel	5000	0.0012	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-sparse-lr1.5-long-s1	funnel	12000	0.0018	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-sparse-lr1.5-long-s2	funnel	12000	0.0018	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-sparse-lr1.5-s1	funnel	5000	0.0018	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-sparse-lr1.5-s2	funnel	5000	0.0018	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e10-sparse-lr2-s1	funnel	5000	0.0024	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e10-sparse-lr2-s2	funnel	5000	0.0024	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2

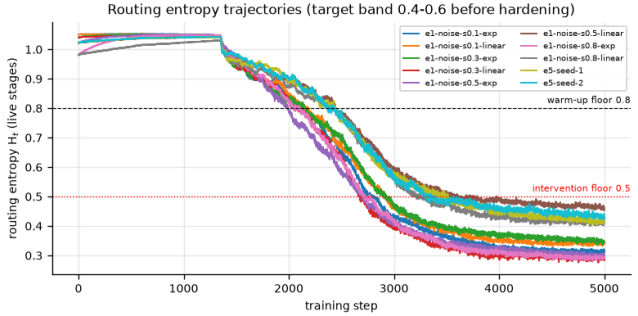
run	variant	steps	lr	warmup_frac	harden_frac	router_freeze_frac	sigma0	noise_decay	T_hi	T_lo	temp_in_B	gumbel	aux_alpha	aux_beta	proxy_weight	entropy_floor	util_ceiling	seed
e2-dense-baseline	dense	5000	0.0006	0.12	0.55	0.35	0.0	const	1.6	0.5	ramp	False	0.01	0.001	0.0	0.5	0.7	0
e3-oracle-capacity-4x	funnel_capacity	5000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e3-oracle-routing-trained	funnel	5000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e3-oracle-tied-experts	funnel_tied	5000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e4-ablate-flat-temp-B	funnel	5000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	flat	False	0.01	0.001	0.5	0.5	0.7	0
e4-ablate-gumbel	funnel	5000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	True	0.01	0.001	0.5	0.5	0.7	0
e4-ablate-instant-hard	funnel	5000	0.0006	0.12	0.0	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e4-ablate-no-aux	funnel	5000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.0	0.0	0.0	0.5	0.7	0
e4-ablate-no-proxy-warmup	funnel	5000	0.0006	0.0	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.0	0.5	0.7	0
e4-ablate-no-router-freeze	funnel	5000	0.0006	0.12	0.55	0.0	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e5-seed-1	funnel	5000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	1
e5-seed-2	funnel	5000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	2
e6-scale-n6000	funnel	5000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e6-scale-n60000	funnel	5000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e7-iso-active	funnel	5000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e8-sparse-d128	funnel	5000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e8-sparse-d160	funnel	5000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e8-sparse-d192	funnel	5000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e8-sparse-d192-wide	funnel	5000	0.0006	0.12	0.55	0.35	0.5	linear	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0
e9-dense-matched-3p5m	dense	5000	0.0006	0.12	0.55	0.35	0.0	const	1.6	0.5	ramp	False	0.01	0.001	0.5	0.5	0.7	0

3. Noise scale x annealing schedule (spec section 3.2)

sigma0	decay	EM(free)	VE	PC	H_final	H_min	wall (s)
0.1	exp	0.522	0.522	0.778	0.316	0.303	261.1
0.1	linear	0.502	0.502	0.768	0.348	0.331	261.1
0.3	exp	0.525	0.525	0.777	0.346	0.336	261.9
0.3	linear	0.520	0.520	0.777	0.290	0.275	262.0
0.5	exp	0.522	0.523	0.784	0.299	0.289	267.1
0.5	linear	0.524	0.524	0.773	0.453	0.453	264.6
0.8	exp	0.524	0.524	0.780	0.291	0.282	265.7
0.8	linear	0.516	0.516	0.773	0.411	0.400	265.9



noise sweep



entropy

3.1 Crossover identification (spec section 3.2)

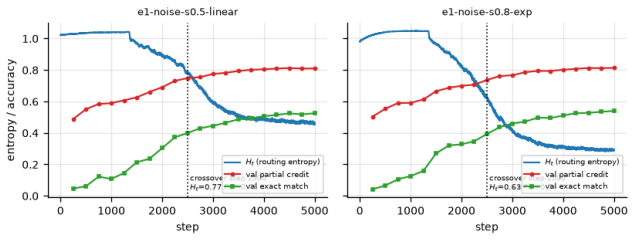
Spec 3.2 asks for the point where routing entropy has stabilised and calculator precision begins climbing to be identified explicitly. Operationally: the first evaluation step at which validation partial credit reaches 90% of its own eventual maximum, together with the entropy at that step. A *late* crossover at *low* entropy is the signature of premature narrowing; an early crossover at high entropy means the router kept exploring while precision improved.

run	crossover step	% of run	H_t at crossover	peak val PC	val EM at peak	H_{min}
e1-noise-s0.1-exp	2750	55%	0.503	0.812	0.528	0.303
e1-noise-s0.1-linear	2500	50%	0.670	0.807	0.523	0.331
e1-noise-s0.3-exp	2500	50%	0.702	0.813	0.539	0.336
e1-noise-s0.3-linear	2750	55%	0.472	0.815	0.535	0.275
e1-noise-s0.5-exp	2750	55%	0.479	0.820	0.543	0.289
e1-noise-s0.5-linear	2500	50%	0.774	0.812	0.525	0.453
e1-noise-s0.8-exp	2500	50%	0.632	0.813	0.540	0.282
e1-noise-s0.8-linear	2750	55%	0.656	0.811	0.530	0.400

Entropy at crossover spans 0.472 (`e1-noise-s0.3-linear`) to 0.774 (`e1-noise-s0.5-linear`). 8 of 8 cells cross only at or after the midpoint of training, and 8 of 8 cross below the spec's 0.8 warm-up floor. Precision therefore arrives largely **after** the routing distribution has narrowed rather than while it is still exploring - the opposite of the failure mode the annealing schedule was designed to prevent, and consistent with noise scale barely moving final accuracy in the table above: by the time precision climbs, the router has largely committed.

4. Routing entropy vs calculator precision (spec section 3.1)

- **e1-noise-s0.1-exp**: peak val partial credit 0.812 at step 4750 (EM 0.528); H_final 0.316, H_min 0.303, guardrail interventions 0.
- **e1-noise-s0.1-linear**: peak val partial credit 0.807 at step 4750 (EM 0.523); H_final 0.348, H_min 0.331, guardrail interventions 0.
- **e1-noise-s0.3-exp**: peak val partial credit 0.813 at step 5000 (EM 0.539); H_final 0.346, H_min 0.336, guardrail interventions 0.
- **e1-noise-s0.3-linear**: peak val partial credit 0.815 at step 4500 (EM 0.535); H_final 0.290, H_min 0.275, guardrail interventions 0.
- **e1-noise-s0.5-exp**: peak val partial credit 0.820 at step 4750 (EM 0.543); H_final 0.299, H_min 0.289, guardrail interventions 0.
- **e1-noise-s0.5-linear**: peak val partial credit 0.812 at step 4500 (EM 0.525); H_final 0.453, H_min 0.453, guardrail interventions 0.
- **e1-noise-s0.8-exp**: peak val partial credit 0.813 at step 5000 (EM 0.540); H_final 0.291, H_min 0.282, guardrail interventions 0.
- **e1-noise-s0.8-linear**: peak val partial credit 0.811 at step 5000 (EM 0.530); H_final 0.411, H_min 0.400, guardrail interventions 0.
- **e5-seed-1**: peak val partial credit 0.814 at step 5000 (EM 0.532); H_final 0.421, H_min 0.404, guardrail interventions 0.
- **e5-seed-2**: peak val partial credit 0.819 at step 5000 (EM 0.537); H_final 0.442, H_min 0.422, guardrail interventions 0.



coupling

4.1 Target-trajectory conformance (spec section 3.1)

The spec asks for `H_t > 0.8` during proxy warm-up, a decay through `0.4-0.6` while soft, hardening once the curve flattens, and an intervention if `H_t < 0.5` before ~step 5k. Measured against that:

run	mean H during warm-up	fraction of trace in 0.4-0.6	first step H<0.5	hardening step	steps with H<0.5 (<5k)	H_final
e1-noise-s0.1-exp	1.052	0.123	2759	2751	2210	0.316
e1-noise-s0.1-linear	1.051	0.137	2901	2751	2084	0.348
e1-noise-s0.3-exp	1.048	0.151	2895	2751	2077	0.346
e1-noise-s0.3-linear	1.044	0.087	2674	2751	2324	0.290
e1-noise-s0.5-exp	1.040	0.119	2695	2751	2302	0.299
e1-noise-s0.5-linear	1.030	0.405	3496	2751	1406	0.453
e1-noise-s0.8-exp	1.023	0.101	2703	2751	2282	0.291
e1-noise-s0.8-linear	0.999	0.419	3183	2751	1762	0.411
e10-sparse-d128-lr0.25-s1	1.031	0.000	-	2751	0	0.666
e10-sparse-d128-lr0.25-s2	1.030	0.000	-	2751	0	0.656
e10-sparse-d128-lr0.5-long-s1	1.031	0.082	6795	6602	0	0.347
e10-sparse-d128-lr0.5-long-s2	1.029	0.168	7003	6602	0	0.373
e10-sparse-d128-lr0.5-s1	1.031	0.409	3354	2751	1582	0.431
e10-sparse-d128-lr0.5-s2	1.029	0.370	4210	2751	412	0.496
e10-sparse-d128-lr1-long-s1	1.031	0.057	6791	6602	0	0.322
e10-sparse-d128-lr1-long-s2	1.029	0.045	6753	6602	0	0.319
e10-sparse-d128-lr1-s1	1.031	0.148	3052	2751	1936	0.352
e10-sparse-d128-lr1-s2	1.029	0.144	3008	2751	1970	0.343
e10-sparse-d128-lr1.5-long-s1	1.032	0.000	-	6602	0	0.915
e10-sparse-d128-lr1.5-long-s2	1.030	0.000	-	6602	0	0.864

run	mean H during warm-up	fraction of trace in 0.4-0.6	first step H<0.5	hardening step	steps with H<0.5 (<5k)	H_final
e10-sparse-d128-lr1.5-s1	1.031	0.371	3827	2751	863	0.471
e10-sparse-d128-lr1.5-s2	1.030	0.401	3440	2751	1458	0.413
e10-sparse-d128-lr2-s1	1.031	0.000	-	2751	0	1.004
e10-sparse-d128-lr2-s2	1.030	0.000	-	2751	0	1.002
e10-sparse-lr0.25-s1	1.031	0.127	-	2751	0	0.588
e10-sparse-lr0.25-s2	1.030	0.094	-	2751	0	0.593
e10-sparse-lr0.5-long-s1	1.031	0.233	7173	6602	0	0.398
e10-sparse-lr0.5-long-s2	1.029	0.207	7140	6602	0	0.384
e10-sparse-lr0.5-s1	1.031	0.250	3167	2751	1782	0.369
e10-sparse-lr0.5-s2	1.029	0.217	3065	2751	1926	0.383
e10-sparse-lr1-long	1.027	0.132	6994	6602	0	0.352
e10-sparse-lr1-long-s1	1.031	0.566	5806	6602	0	0.417
e10-sparse-lr1-long-s2	1.029	0.240	7122	6602	0	0.378
e10-sparse-lr1-s1	1.031	0.328	3237	2751	1742	0.387
e10-sparse-lr1-s2	1.029	0.391	3109	2751	1864	0.400
e10-sparse-lr1.5-long-s1	1.032	0.074	-	6602	0	0.512
e10-sparse-lr1.5-long-s2	1.032	0.019	6448	6602	0	0.472
e10-sparse-lr1.5-s1	1.031	0.000	-	2751	0	0.768
e10-sparse-lr1.5-s2	1.029	0.000	-	2751	0	0.653
e10-sparse-lr2-s1	1.032	0.118	-	2751	0	0.542
e10-sparse-lr2-s2	1.031	0.000	-	2751	0	0.706

run	mean H during warm-up	fraction of trace in 0.4-0.6	first step H<0.5	hardening step	steps with H<0.5 (<5k)	H_final
e3-oracle-capacity-4x	1.030	0.190	2988	2751	1957	0.360
e3-oracle-tied-experts	1.029	0.135	-	2751	0	0.583
e4-ablate-flat-temp-B	1.030	0.450	2943	2751	2034	0.413
e4-ablate-gumbel	1.015	0.434	3088	2751	1896	0.400
e4-ablate-instant-hard	1.046	0.094	1107	601	3874	0.140
e4-ablate-no-aux	1.029	0.079	2028	2751	2967	0.188
e4-ablate-no-proxy-warmup	n/a (no warm-up phase)	0.089	1699	2751	3276	0.191
e4-ablate-no-router-freeze	1.030	0.133	2870	2751	2110	0.349
e5-seed-1	1.031	0.423	3353	2751	1514	0.421
e5-seed-2	1.032	0.417	3288	2751	1679	0.442
e6-scale-n6000	1.030	0.160	2720	2751	2271	0.369
e6-scale-n60000	1.030	0.461	3010	2751	1945	0.402
e7-iso-active	1.031	0.454	3089	2751	1894	0.439
e8-sparse-d128	1.028	0.400	3544	2751	1411	0.450
e8-sparse-d160	1.031	0.424	3131	2751	1855	0.404
e8-sparse-d192	1.027	0.130	2934	2751	2060	0.347
e8-sparse-d192-wide	1.028	0.414	3222	2751	1747	0.403

Deviation, stated explicitly. The absolute rule 'intervene if `H_t < 0.5` before step ~5k' contradicts the same paragraph's instruction to *harden once the curve flattens*: hardening is configured at steps 601-6602 in these runs, and its entire purpose is to drive `H_t` toward 0. Applying the rule literally over the whole run therefore fires on every intended hardening (column 'steps with H<0.5 (<5k)'). The guardrail window used here is the first 30% of training - strictly *before* the hardening point - which preserves the rule's intent (catch premature collapse) without flagging the designed transition. Under that window no run needed an intervention; under the strict reading every run above would have triggered hundreds of times: that column is nonzero for all of them and is generated by the *intended* hardening at step ~2751, not by a collapse.

Mean fraction of each trace inside the 0.4-0.6 target band: 0.20 (range 0.00-0.57). The band is crossed during phase B in every run but is a transient, not a plateau - at this parameter scale the joint noise/temperature anneal passes through it in roughly 357 steps.

5. Oracle bounds and gaps (spec section 5.1)

Oracle definitions used here (spec section 5.1 leaves the exact protocol open; this is what was implemented):

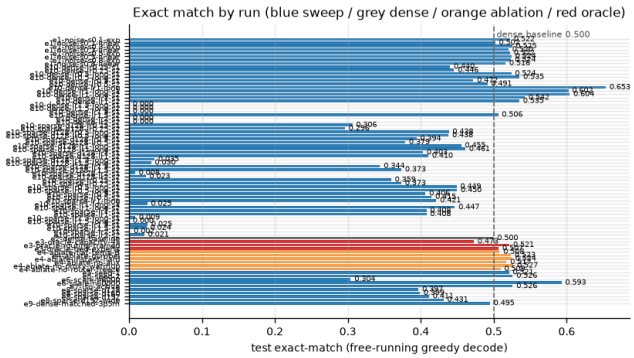
- `e3-oracle-routing-trained` - trained with ground-truth `[3,2,2]` paths forced at every stage, so tiles specialise on the *correct* partition. Isolates routing inefficiency from tile learning.

- `test_free_gt_routing` (same weights) - the learned model re-evaluated with true paths injected at test time; the delta against self-routing is the pure routing cost with capacity and weights held fixed.
- `e3-oracle-capacity-4x` - capacity oracle sub-variant (1): 4x expert inner width, **deliberately over the 10M budget**, to see what tile capacity buys.
- `e3-oracle-tied-experts` - not an upper bound but a *capacity floor*: one shared expert pool across all stages (weight tying) gives strictly less effective capacity inside the same budget, so it brackets the capacity axis from below while the 4x oracle brackets it from above.
- The 10M-budget actual-tile-size case is simply the routed model itself.

`gap = (oracle - model) / oracle`; `headroom = (model - dense) / (oracle - dense)`.

oracle variant	EM oracle	PC oracle	EM model	EM dense	gap	headroom captured
e3-oracle-capacity-4x	0.473	0.739	0.524	0.500	-0.108	-0.889
e3-oracle-routing-trained	0.521	0.775	0.524	0.500	-0.006	1.143
e3-oracle-tied-experts	0.507	0.769	0.524	0.500	-0.034	3.429

For the trained routing oracle `e3-oracle-routing-trained`, forcing ground-truth `[3,2,2]` paths on the *same learned weights* gives EM 0.521 vs self-routed 0.521: that difference is the pure routing cost, with tile capacity held fixed.



oracle gaps

6. Collapse-mitigation ablations (spec section 4.3)

ablation	EM(free)	PC	H_final	H_min	guardrail hits
e4-ablate-flat-temp-B	0.508	0.767	0.413	0.405	0
e4-ablate-gumbel	0.523	0.780	0.400	0.396	0
e4-ablate-instant-hard	0.524	0.777	0.140	0.127	14
e4-ablate-no-aux	0.517	0.775	0.188	0.179	2
e4-ablate-no-proxy-warmup	0.527	0.779	0.191	0.181	7
e4-ablate-no-router-freeze	0.509	0.768	0.349	0.340	0

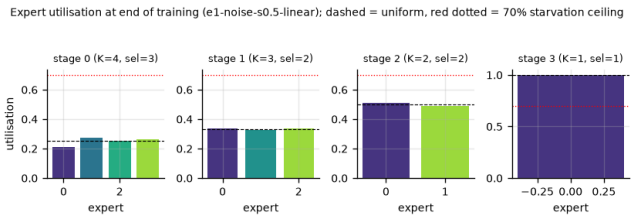
7. Stability guardrails and expert utilisation (spec section 6.2)

run	interventions	first trigger steps	H_min	H_final	non-finite steps skipped
e1-noise-s0.1-exp	0	-	0.303	0.316	0
e1-noise-s0.1-linear	0	-	0.331	0.348	0

run	interventions	first trigger steps	H_min	H_final	non-finite steps skipped
e1-noise-s0.3-exp	0	-	0.336	0.346	0
e1-noise-s0.3-linear	0	-	0.275	0.290	0
e1-noise-s0.5-exp	0	-	0.289	0.299	0
e1-noise-s0.5-linear	0	-	0.453	0.453	0
e1-noise-s0.8-exp	0	-	0.282	0.291	0
e1-noise-s0.8-linear	0	-	0.400	0.411	0
e10-dense-lr0.25-s1	0	-	-	-	0
e10-dense-lr0.25-s2	0	-	-	-	0
e10-dense-lr0.5-long-s1	0	-	-	-	0
e10-dense-lr0.5-long-s2	0	-	-	-	0
e10-dense-lr0.5-s1	0	-	-	-	0
e10-dense-lr0.5-s2	0	-	-	-	0
e10-dense-lr1-long	0	-	-	-	0
e10-dense-lr1-long-s1	0	-	-	-	0
e10-dense-lr1-long-s2	0	-	-	-	0
e10-dense-lr1-s1	0	-	-	-	0
e10-dense-lr1-s2	0	-	-	-	0
e10-dense-lr1.5-long-s1	0	-	-	-	9353
e10-dense-lr1.5-long-s2	0	-	-	-	8943
e10-dense-lr1.5-s1	0	-	-	-	2503
e10-dense-lr1.5-s2	0	-	-	-	0
e10-dense-lr2-s1	0	-	-	-	4014
e10-dense-lr2-s2	0	-	-	-	3808
e10-sparse-d128-lr0.25-s1	0	-	0.661	0.666	0
e10-sparse-d128-lr0.25-s2	0	-	0.651	0.656	0
e10-sparse-d128-lr0.5-long-s1	0	-	0.332	0.347	0
e10-sparse-d128-lr0.5-long-s2	0	-	0.363	0.373	0
e10-sparse-d128-lr0.5-s1	0	-	0.423	0.431	0
e10-sparse-d128-lr0.5-s2	0	-	0.487	0.496	0
e10-sparse-d128-lr1-long-s1	0	-	0.315	0.322	0

run	interventions	first trigger steps	H_min	H_final	non-finite steps skipped
e10-sparse-d128-lr1-long-s2	0	-	0.316	0.319	0
e10-sparse-d128-lr1-s1	0	-	0.336	0.352	0
e10-sparse-d128-lr1-s2	0	-	0.328	0.343	0
e10-sparse-d128-lr1.5-long-s1	0	-	0.884	0.915	0
e10-sparse-d128-lr1.5-long-s2	0	-	0.827	0.864	0
e10-sparse-d128-lr1.5-s1	0	-	0.463	0.471	0
e10-sparse-d128-lr1.5-s2	0	-	0.409	0.413	0
e10-sparse-d128-lr2-s1	0	-	0.994	1.004	2392
e10-sparse-d128-lr2-s2	0	-	0.962	1.002	0
e10-sparse-lr0.25-s1	0	-	0.581	0.588	0
e10-sparse-lr0.25-s2	0	-	0.588	0.593	0
e10-sparse-lr0.5-long-s1	0	-	0.374	0.398	0
e10-sparse-lr0.5-long-s2	0	-	0.371	0.384	0
e10-sparse-lr0.5-s1	0	-	0.364	0.369	0
e10-sparse-lr0.5-s2	0	-	0.369	0.383	0
e10-sparse-lr1-long	0	-	0.333	0.352	0
e10-sparse-lr1-long-s1	1	3517	0.405	0.417	0
e10-sparse-lr1-long-s2	0	-	0.367	0.378	0
e10-sparse-lr1-s1	0	-	0.381	0.387	0
e10-sparse-lr1-s2	0	-	0.390	0.400	0
e10-sparse-lr1.5-long-s1	0	-	0.512	0.512	6309
e10-sparse-lr1.5-long-s2	0	-	0.471	0.472	7635
e10-sparse-lr1.5-s1	0	-	0.735	0.768	0
e10-sparse-lr1.5-s2	0	-	0.610	0.653	0
e10-sparse-lr2-s1	0	-	0.539	0.542	2007
e10-sparse-lr2-s2	0	-	0.673	0.706	0
e2-dense-baseline	0	-	-	-	0
e3-oracle-capacity-4x	0	-	0.345	0.360	0
e3-oracle-routing-trained	0	-	-	-	0
e3-oracle-tied-experts	0	-	0.559	0.583	0

run	interventions	first trigger steps	H_min	H_final	non-finite steps skipped
e4-ablate-flat-temp-B	0	-	0.405	0.413	0
e4-ablate-gumbel	0	-	0.396	0.400	0
e4-ablate-instant-hard	14	676, 826, 894, 945, 996, 1047 …	0.127	0.140	0
e4-ablate-no-aux	2	1368, 2152	0.179	0.188	0
e4-ablate-no-proxy-warmup	7	1086, 1222, 1273, 1324, 1375, 1426 …	0.181	0.191	0
e4-ablate-no-router-freeze	0	-	0.340	0.349	0
e5-seed-1	0	-	0.404	0.421	0
e5-seed-2	0	-	0.422	0.442	0
e6-scale-n6000	0	-	0.353	0.369	0
e6-scale-n60000	0	-	0.394	0.402	0
e7-iso-active	0	-	0.421	0.439	0
e8-sparse-d128	0	-	0.449	0.450	0
e8-sparse-d160	0	-	0.398	0.404	0
e8-sparse-d192	0	-	0.340	0.347	0
e8-sparse-d192-wide	0	-	0.399	0.403	0
e9-dense-matched-3p5m	0	-	-	-	0



utilisation

8. Per-domain / difficulty breakdown

e1-noise-s0.5-linear

domain	tier	n	EM	VE
0	1	667	0.999	0.999
0	2	667	0.633	0.633
0	3	667	0.168	0.168
1	1	666	1.000	1.000
1	2	666	0.213	0.213
1	3	666	0.006	0.006
2	1	667	0.906	0.906
2	2	667	0.067	0.067
2	3	667	0.727	0.727

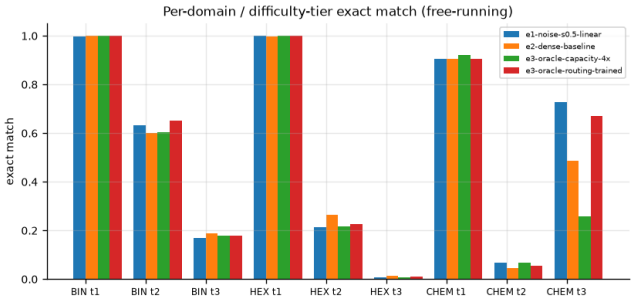
e2-dense-baseline

domain	tier	n	EM	VE
0	1	667	1.000	1.000
0	2	667	0.600	0.600

domain	tier	n	EM	VE
0	3	667	0.187	0.187
1	1	666	0.997	0.997
1	2	666	0.264	0.264
1	3	666	0.014	0.014
2	1	667	0.904	0.904
2	2	667	0.045	0.045
2	3	667	0.486	0.486

e3-oracle-capacity-4x

domain	tier	n	EM	VE
0	1	667	1.000	1.000
0	2	667	0.604	0.604
0	3	667	0.180	0.180
1	1	666	1.000	1.000
1	2	666	0.216	0.216
1	3	666	0.008	0.008
2	1	667	0.922	0.922
2	2	667	0.066	0.066
2	3	667	0.258	0.258



domains

9. Training cost vs dense baseline

run	wall (s)	ratio to dense	steps
e1-noise-s0.1-exp	261.1	1.12	5000
e1-noise-s0.1-linear	261.1	1.12	5000
e1-noise-s0.3-exp	261.9	1.13	5000
e1-noise-s0.3-linear	262.0	1.13	5000
e1-noise-s0.5-exp	267.1	1.15	5000
e1-noise-s0.5-linear	264.6	1.14	5000
e1-noise-s0.8-exp	265.7	1.14	5000
e1-noise-s0.8-linear	265.9	1.14	5000
e10-dense-lr0.25-s1	128.4	0.55	5000
e10-dense-lr0.25-s2	128.3	0.55	5000
e10-dense-lr0.5-long-s1	311.9	1.34	12000
e10-dense-lr0.5-long-s2	311.4	1.34	12000
e10-dense-lr0.5-s1	136.8	0.59	5000
e10-dense-lr0.5-s2	131.7	0.57	5000
e10-dense-lr1-long	314.1	1.35	12000
e10-dense-lr1-long-s1	309.1	1.33	12000
e10-dense-lr1-long-s2	313.9	1.35	12000
e10-dense-lr1-s1	129.2	0.56	5000
e10-dense-lr1-s2	131.6	0.57	5000
e10-dense-lr1.5-long-s1	284.2	1.22	12000
e10-dense-lr1.5-long-s2	283.7	1.22	12000
e10-dense-lr1.5-s1	121.8	0.52	5000

run	wall (s)	ratio to dense	steps
e10-dense-lr1.5-s2	129.0	0.56	5000
e10-dense-lr2-s1	115.8	0.50	5000
e10-dense-lr2-s2	117.7	0.51	5000
e10-sparse-d128-lr0.25-s1	137.0	0.59	5000
e10-sparse-d128-lr0.25-s2	137.4	0.59	5000
e10-sparse-d128-lr0.5-long-s1	327.5	1.41	12000
e10-sparse-d128-lr0.5-long-s2	327.9	1.41	12000
e10-sparse-d128-lr0.5-s1	136.7	0.59	5000
e10-sparse-d128-lr0.5-s2	136.3	0.59	5000
e10-sparse-d128-lr1-long-s1	328.7	1.41	12000
e10-sparse-d128-lr1-long-s2	329.1	1.42	12000
e10-sparse-d128-lr1-s1	136.7	0.59	5000
e10-sparse-d128-lr1-s2	137.2	0.59	5000
e10-sparse-d128-lr1.5-long-s1	325.5	1.40	12000
e10-sparse-d128-lr1.5-long-s2	328.6	1.41	12000
e10-sparse-d128-lr1.5-s1	137.2	0.59	5000
e10-sparse-d128-lr1.5-s2	136.8	0.59	5000
e10-sparse-d128-lr2-s1	126.9	0.55	5000
e10-sparse-d128-lr2-s2	345.6	1.49	5000
e10-sparse-lr0.25-s1	137.6	0.59	5000
e10-sparse-lr0.25-s2	136.6	0.59	5000
e10-sparse-lr0.5-long-s1	334.5	1.44	12000
e10-sparse-lr0.5-long-s2	342.3	1.47	12000
e10-sparse-lr0.5-s1	137.7	0.59	5000
e10-sparse-lr0.5-s2	136.7	0.59	5000
e10-sparse-lr1-long	330.5	1.42	12000
e10-sparse-lr1-long-s1	329.9	1.42	12000
e10-sparse-lr1-long-s2	330.2	1.42	12000
e10-sparse-lr1-s1	139.0	0.60	5000
e10-sparse-lr1-s2	137.8	0.59	5000
e10-sparse-lr1.5-long-s1	309.4	1.33	12000
e10-sparse-lr1.5-long-s2	301.8	1.30	12000
e10-sparse-lr1.5-s1	135.9	0.58	5000
e10-sparse-lr1.5-s2	137.4	0.59	5000
e10-sparse-lr2-s1	131.1	0.56	5000
e10-sparse-lr2-s2	142.9	0.61	5000
e2-dense-baseline	232.4	1.00	5000
e3-oracle-capacity-4x	302.9	1.30	5000
e3-oracle-routing-trained	249.5	1.07	5000
e3-oracle-tied-experts	267.0	1.15	5000
e4-ablate-flat-temp-B	266.2	1.15	5000
e4-ablate-gumbel	266.7	1.15	5000
e4-ablate-instant-hard	267.2	1.15	5000
e4-ablate-no-aux	266.1	1.15	5000
e4-ablate-no-proxy-warmup	266.4	1.15	5000
e4-ablate-no-router-freeze	266.6	1.15	5000
e5-seed-1	266.2	1.15	5000
e5-seed-2	266.0	1.14	5000
e6-scale-n6000	257.1	1.11	5000
e6-scale-n60000	270.6	1.16	5000
e7-iso-active	268.7	1.16	5000
e8-sparse-d128	136.1	0.59	5000
e8-sparse-d160	144.7	0.62	5000
e8-sparse-d192	138.6	0.60	5000
e8-sparse-d192-wide	159.1	0.68	5000
e9-dense-matched-3p5m	130.4	0.56	5000



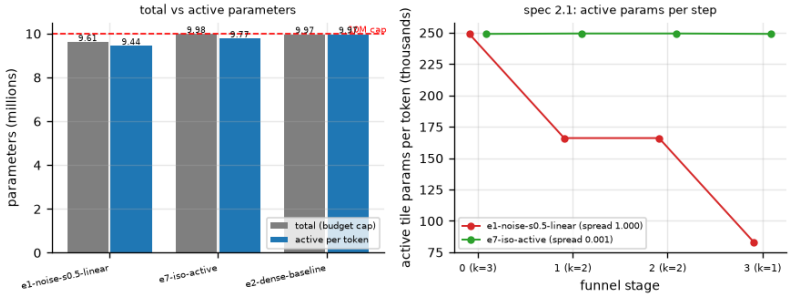
loss trajectories

10. Active parameters per token vs the dense baseline

run	variant	total params	active/ token	active / dense total	skipped/ token	% skipped	active tiles	all tiles	tile use	per-stage spread	~constant per step?
e1-noise-s0.1-exp	funnel	9,607,384	9,441,368	0.947	166,016	1.73	664,064	830,080	0.800	1.0000	**NO**
e1-noise-s0.1-linear	funnel	9,607,384	9,441,368	0.947	166,016	1.73	664,064	830,080	0.800	1.0000	**NO**
e1-noise-s0.3-exp	funnel	9,607,384	9,441,368	0.947	166,016	1.73	664,064	830,080	0.800	1.0000	**NO**
e1-noise-s0.3-linear	funnel	9,607,384	9,441,368	0.947	166,016	1.73	664,064	830,080	0.800	1.0000	**NO**
e1-noise-s0.5-exp	funnel	9,607,384	9,441,368	0.947	166,016	1.73	664,064	830,080	0.800	1.0000	**NO**
e1-noise-s0.5-linear	funnel	9,607,384	9,441,368	0.947	166,016	1.73	664,064	830,080	0.800	1.0000	**NO**
e1-noise-s0.8-exp	funnel	9,607,384	9,441,368	0.947	166,016	1.73	664,064	830,080	0.800	1.0000	**NO**
e1-noise-s0.8-linear	funnel	9,607,384	9,441,368	0.947	166,016	1.73	664,064	830,080	0.800	1.0000	**NO**
e10-sparse-d128-lr0.25-s1	funnel	3,514,943	3,014,824	0.303	500,119	14.23	2,400,368	2,900,487	0.828	0.0004	yes
e10-sparse-d128-lr0.25-s2	funnel	3,514,943	3,014,824	0.303	500,119	14.23	2,400,368	2,900,487	0.828	0.0004	yes
e10-sparse-d128-lr0.5-long-s1	funnel	3,514,943	3,014,824	0.303	500,119	14.23	2,400,368	2,900,487	0.828	0.0004	yes
e10-sparse-d128-lr0.5-long-s2	funnel	3,514,943	3,014,824	0.303	500,119	14.23	2,400,368	2,900,487	0.828	0.0004	yes
e10-sparse-d128-lr0.5-s1	funnel	3,514,943	3,014,824	0.303	500,119	14.23	2,400,368	2,900,487	0.828	0.0004	yes
e10-sparse-d128-lr0.5-s2	funnel	3,514,943	3,014,824	0.303	500,119	14.23	2,400,368	2,900,487	0.828	0.0004	yes
e10-sparse-d128-lr1-long-s1	funnel	3,514,943	3,014,824	0.303	500,119	14.23	2,400,368	2,900,487	0.828	0.0004	yes
e10-sparse-d128-lr1-long-s2	funnel	3,514,943	3,014,824	0.303	500,119	14.23	2,400,368	2,900,487	0.828	0.0004	yes
e10-sparse-d128-lr1-s1	funnel	3,514,943	3,014,824	0.303	500,119	14.23	2,400,368	2,900,487	0.828	0.0004	yes
e10-sparse-d128-lr1-s2	funnel	3,514,943	3,014,824	0.303	500,119	14.23	2,400,368	2,900,487	0.828	0.0004	yes
e10-sparse-d128-lr1.5-long-s1	funnel	3,514,943	3,014,824	0.303	500,119	14.23	2,400,368	2,900,487	0.828	0.0004	yes
e10-sparse-d128-lr1.5-long-s2	funnel	3,514,943	3,014,824	0.303	500,119	14.23	2,400,368	2,900,487	0.828	0.0004	yes
e10-sparse-d128-lr1.5-s1	funnel	3,514,943	3,014,824	0.303	500,119	14.23	2,400,368	2,900,487	0.828	0.0004	yes
e10-sparse-d128-lr1.5-s2	funnel	3,514,943	3,014,824	0.303	500,119	14.23	2,400,368	2,900,487	0.828	0.0004	yes
e10-sparse-d128-lr2-s1	funnel	3,514,943	3,014,824	0.303	500,119	14.23	2,400,368	2,900,487	0.828	0.0004	yes
e10-sparse-d128-lr2-s2	funnel	3,514,943	3,014,824	0.303	500,119	14.23	2,400,368	2,900,487	0.828	0.0004	yes
e10-sparse-lr0.25-s1	funnel	3,677,355	3,243,848	0.326	433,507	11.79	2,080,528	2,514,035	0.828	0.0007	yes
e10-sparse-lr0.25-s2	funnel	3,677,355	3,243,848	0.326	433,507	11.79	2,080,528	2,514,035	0.828	0.0007	yes
e10-sparse-lr0.5-long-s1	funnel	3,677,355	3,243,848	0.326	433,507	11.79	2,080,528	2,514,035	0.828	0.0007	yes
e10-sparse-lr0.5-long-s2	funnel	3,677,355	3,243,848	0.326	433,507	11.79	2,080,528	2,514,035	0.828	0.0007	yes
e10-sparse-lr0.5-s1	funnel	3,677,355	3,243,848	0.326	433,507	11.79	2,080,528	2,514,035	0.828	0.0007	yes
e10-sparse-lr0.5-s2	funnel	3,677,355	3,243,848	0.326	433,507	11.79	2,080,528	2,514,035	0.828	0.0007	yes
e10-sparse-lr1-long	funnel	3,677,355	3,243,848	0.326	433,507	11.79	2,080,528	2,514,035	0.828	0.0007	yes
e10-sparse-lr1-long-s1	funnel	3,677,355	3,243,848	0.326	433,507	11.79	2,080,528	2,514,035	0.828	0.0007	yes
e10-sparse-lr1-long-s2	funnel	3,677,355	3,243,848	0.326	433,507	11.79	2,080,528	2,514,035	0.828	0.0007	yes
e10-sparse-lr1-s1	funnel	3,677,355	3,243,848	0.326	433,507	11.79	2,080,528	2,514,035	0.828	0.0007	yes
e10-sparse-lr1-s2	funnel	3,677,355	3,243,848	0.326	433,507	11.79	2,080,528	2,514,035	0.828	0.0007	yes
e10-sparse-lr1.5-long-s1	funnel	3,677,355	3,243,848	0.326	433,507	11.79	2,080,528	2,514,035	0.828	0.0007	yes
e10-sparse-lr1.5-long-s2	funnel	3,677,355	3,243,848	0.326	433,507	11.79	2,080,528	2,514,035	0.828	0.0007	yes
e10-sparse-lr1.5-s1	funnel	3,677,355	3,243,848	0.326	433,507	11.79	2,080,528	2,514,035	0.828	0.0007	yes
e10-sparse-lr1.5-s2	funnel	3,677,355	3,243,848	0.326	433,507	11.79	2,080,528	2,514,035	0.828	0.0007	yes
e10-sparse-lr2-s1	funnel	3,677,355	3,243,848	0.326	433,507	11.79	2,080,528	2,514,035	0.828	0.0007	yes
e10-sparse-lr2-s2	funnel	3,677,355	3,243,848	0.326	433,507	11.79	2,080,528	2,514,035	0.828	0.0007	yes
e2-dense-baseline	dense	9,965,120	9,965,120	1.000	0	0.00	0	0	-	0.0000	yes
e3-oracle-capacity-4x	funnel_capacity	12,068,824	11,410,520	1.145	658,304	5.45	2,633,216	3,291,520	0.800	1.0000	**NO**
e3-oracle-routing-trained	funnel	9,607,384	9,441,368	0.947	166,016	1.73	664,064	830,080	0.800	1.0000	**NO**
e3-oracle-tied-experts	funnel_tied	9,109,336	8,943,320	0.897	166,016	1.82	664,064	830,080	0.800	1.0000	**NO**
e4-ablate-flat-temp-B	funnel	9,607,384	9,441,368	0.947	166,016	1.73	664,064	830,080	0.800	1.0000	**NO**
e4-ablate-gumbel	funnel	9,607,384	9,441,368	0.947	166,016	1.73	664,064	830,080	0.800	1.0000	**NO**
e4-ablate-instant-hard	funnel	9,607,384	9,441,368	0.947	166,016	1.73	664,064	830,080	0.800	1.0000	**NO**
e4-ablate-no-aux	funnel	9,607,384	9,441,368	0.947	166,016	1.73	664,064	830,080	0.800	1.0000	**NO**

run	variant	total params	active/ token	active / dense total	skipped/ token	% skipped	active tiles	all tiles	tile use	per-stage spread	~constant per step?
e4-ablate-no-proxy-warmup	funnel	9,607,384	9,441,368	0.947	166,016	1.73	664,064	830,080	0.800	1.0000	**NO**
e4-ablate-no-router-freeze	funnel	9,607,384	9,441,368	0.947	166,016	1.73	664,064	830,080	0.800	1.0000	**NO**
e5-seed-1	funnel	9,607,384	9,441,368	0.947	166,016	1.73	664,064	830,080	0.800	1.0000	**NO**
e5-seed-2	funnel	9,607,384	9,441,368	0.947	166,016	1.73	664,064	830,080	0.800	1.0000	**NO**
e6-scale-n6000	funnel	9,607,384	9,441,368	0.947	166,016	1.73	664,064	830,080	0.800	1.0000	**NO**
e6-scale-n60000	funnel	9,607,384	9,441,368	0.947	166,016	1.73	664,064	830,080	0.800	1.0000	**NO**
e7-iso-active	funnel	9,981,728	9,774,047	0.981	207,681	2.08	996,743	1,204,424	0.828	0.0013	yes
e8-sparse-d128	funnel	3,514,943	3,014,824	0.303	500,119	14.23	2,400,368	2,900,487	0.828	0.0004	yes
e8-sparse-d160	funnel	3,727,477	3,227,362	0.324	500,115	13.42	2,400,426	2,900,541	0.828	0.0008	yes
e8-sparse-d192	funnel	3,677,355	3,243,848	0.326	433,507	11.79	2,080,528	2,514,035	0.828	0.0007	yes
e8-sparse-d192-wide	funnel	5,028,705	4,362,273	0.438	666,432	13.25	3,198,953	3,865,385	0.828	0.0002	yes
e9-dense-matched-3p5m	dense	3,505,536	3,505,536	0.352	0	0.00	0	0	-	0.0000	yes

`active/token` counts parameters that participate in producing one token under hard `[3,2,2,1]` routing. Only expert tiles are switchable; the trunk (embeddings, encoder/decoder blocks, per-stage bodies, attention, readout head, router/gates) executes for every token in every configuration and is always counted in full. `active / dense total` divides by the dense baseline's total, which *is* its per-token cost - that ratio is the honest inference comparison. `tile use` is active tiles / all tiles. Spread is (max-min)/mean of per-stage active tile params; the spec asks for it to be small.



Per-stage active budget (spec 2.1 conformance)

run	stage widths	selects	active tile params per stage (/ separated)	spread (max-min)/ mean	roughly constant?
e1-noise-s0.1-exp	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 166016 / 166016 / 83008	1.0000	**NO**
e1-noise-s0.1-linear	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 166016 / 166016 / 83008	1.0000	**NO**
e1-noise-s0.3-exp	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 166016 / 166016 / 83008	1.0000	**NO**
e1-noise-s0.3-linear	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 166016 / 166016 / 83008	1.0000	**NO**
e1-noise-s0.5-exp	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 166016 / 166016 / 83008	1.0000	**NO**
e1-noise-s0.5-linear	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 166016 / 166016 / 83008	1.0000	**NO**
e1-noise-s0.8-exp	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 166016 / 166016 / 83008	1.0000	**NO**
e1-noise-s0.8-linear	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 166016 / 166016 / 83008	1.0000	**NO**
e10-sparse-d128-lr0.25-s1	[4, 3, 2, 1]	[3, 2, 2, 1]	600219 / 600092 / 600092 / 599965	0.0004	yes
e10-sparse-d128-lr0.25-s2	[4, 3, 2, 1]	[3, 2, 2, 1]	600219 / 600092 / 600092 / 599965	0.0004	yes
e10-sparse-d128-lr0.5-long-s1	[4, 3, 2, 1]	[3, 2, 2, 1]	600219 / 600092 / 600092 / 599965	0.0004	yes
e10-sparse-d128-lr0.5-long-s2	[4, 3, 2, 1]	[3, 2, 2, 1]	600219 / 600092 / 600092 / 599965	0.0004	yes
e10-sparse-d128-lr0.5-s1	[4, 3, 2, 1]	[3, 2, 2, 1]	600219 / 600092 / 600092 / 599965	0.0004	yes
e10-sparse-d128-lr0.5-s2	[4, 3, 2, 1]	[3, 2, 2, 1]	600219 / 600092 / 600092 / 599965	0.0004	yes
e10-sparse-d128-lr1-long-s1	[4, 3, 2, 1]	[3, 2, 2, 1]	600219 / 600092 / 600092 / 599965	0.0004	yes
e10-sparse-d128-lr1-long-s2	[4, 3, 2, 1]	[3, 2, 2, 1]	600219 / 600092 / 600092 / 599965	0.0004	yes
e10-sparse-d128-lr1-s1	[4, 3, 2, 1]	[3, 2, 2, 1]	600219 / 600092 / 600092 / 599965	0.0004	yes
e10-sparse-d128-lr1-s2	[4, 3, 2, 1]	[3, 2, 2, 1]	600219 / 600092 / 600092 / 599965	0.0004	yes
e10-sparse-d128-lr1.5-long-s1	[4, 3, 2, 1]	[3, 2, 2, 1]	600219 / 600092 / 600092 / 599965	0.0004	yes
e10-sparse-d128-lr1.5-long-s2	[4, 3, 2, 1]	[3, 2, 2, 1]	600219 / 600092 / 600092 / 599965	0.0004	yes
e10-sparse-d128-lr1.5-s1	[4, 3, 2, 1]	[3, 2, 2, 1]	600219 / 600092 / 600092 / 599965	0.0004	yes
e10-sparse-d128-lr1.5-s2	[4, 3, 2, 1]	[3, 2, 2, 1]	600219 / 600092 / 600092 / 599965	0.0004	yes
e10-sparse-d128-lr2-s1	[4, 3, 2, 1]	[3, 2, 2, 1]	600219 / 600092 / 600092 / 599965	0.0004	yes
e10-sparse-d128-lr2-s2	[4, 3, 2, 1]	[3, 2, 2, 1]	600219 / 600092 / 600092 / 599965	0.0004	yes
e10-sparse-lr0.25-s1	[4, 3, 2, 1]	[3, 2, 2, 1]	520323 / 520132 / 520132 / 519941	0.0007	yes
e10-sparse-lr0.25-s2	[4, 3, 2, 1]	[3, 2, 2, 1]	520323 / 520132 / 520132 / 519941	0.0007	yes
e10-sparse-lr0.5-long-s1	[4, 3, 2, 1]	[3, 2, 2, 1]	520323 / 520132 / 520132 / 519941	0.0007	yes
e10-sparse-lr0.5-long-s2	[4, 3, 2, 1]	[3, 2, 2, 1]	520323 / 520132 / 520132 / 519941	0.0007	yes
e10-sparse-lr0.5-s1	[4, 3, 2, 1]	[3, 2, 2, 1]	520323 / 520132 / 520132 / 519941	0.0007	yes
e10-sparse-lr0.5-s2	[4, 3, 2, 1]	[3, 2, 2, 1]	520323 / 520132 / 520132 / 519941	0.0007	yes
e10-sparse-lr1-long	[4, 3, 2, 1]	[3, 2, 2, 1]	520323 / 520132 / 520132 / 519941	0.0007	yes
e10-sparse-lr1-long-s1	[4, 3, 2, 1]	[3, 2, 2, 1]	520323 / 520132 / 520132 / 519941	0.0007	yes
e10-sparse-lr1-long-s2	[4, 3, 2, 1]	[3, 2, 2, 1]	520323 / 520132 / 520132 / 519941	0.0007	yes
e10-sparse-lr1-s1	[4, 3, 2, 1]	[3, 2, 2, 1]	520323 / 520132 / 520132 / 519941	0.0007	yes
e10-sparse-lr1-s2	[4, 3, 2, 1]	[3, 2, 2, 1]	520323 / 520132 / 520132 / 519941	0.0007	yes
e10-sparse-lr1.5-long-s1	[4, 3, 2, 1]	[3, 2, 2, 1]	520323 / 520132 / 520132 / 519941	0.0007	yes
e10-sparse-lr1.5-long-s2	[4, 3, 2, 1]	[3, 2, 2, 1]	520323 / 520132 / 520132 / 519941	0.0007	yes
e10-sparse-lr1.5-s1	[4, 3, 2, 1]	[3, 2, 2, 1]	520323 / 520132 / 520132 / 519941	0.0007	yes
e10-sparse-lr1.5-s2	[4, 3, 2, 1]	[3, 2, 2, 1]	520323 / 520132 / 520132 / 519941	0.0007	yes
e10-sparse-lr2-s1	[4, 3, 2, 1]	[3, 2, 2, 1]	520323 / 520132 / 520132 / 519941	0.0007	yes
e10-sparse-lr2-s2	[4, 3, 2, 1]	[3, 2, 2, 1]	520323 / 520132 / 520132 / 519941	0.0007	yes
e3-oracle-capacity-4x	[4, 3, 2, 1]	[3, 2, 2, 1]	987456 / 658304 / 658304 / 329152	1.0000	**NO**
e3-oracle-routing-trained	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 166016 / 166016 / 83008	1.0000	**NO**
e3-oracle-tied-experts	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 166016 / 166016 / 83008	1.0000	**NO**
e4-ablate-flat-temp-B	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 166016 / 166016 / 83008	1.0000	**NO**
e4-ablate-gumbel	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 166016 / 166016 / 83008	1.0000	**NO**
e4-ablate-instant-hard	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 166016 / 166016 / 83008	1.0000	**NO**
e4-ablate-no-aux	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 166016 / 166016 / 83008	1.0000	**NO**

run	stage widths	selects	active tile params per stage (/ separated)	spread (max-min)/ mean	roughly constant?
e4-ablate-no-proxy-warmup	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 166016 / 166016 / 83008	1.0000	**NO**
e4-ablate-no-router-freeze	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 166016 / 166016 / 83008	1.0000	**NO**
e5-seed-1	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 166016 / 166016 / 83008	1.0000	**NO**
e5-seed-2	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 166016 / 166016 / 83008	1.0000	**NO**
e6-scale-n6000	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 166016 / 166016 / 83008	1.0000	**NO**
e6-scale-n60000	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 166016 / 166016 / 83008	1.0000	**NO**
e7-iso-active	[4, 3, 2, 1]	[3, 2, 2, 1]	249024 / 249346 / 249346 / 249027	0.0013	yes
e8-sparse-d128	[4, 3, 2, 1]	[3, 2, 2, 1]	600219 / 600092 / 600092 / 599965	0.0004	yes
e8-sparse-d160	[4, 3, 2, 1]	[3, 2, 2, 1]	600426 / 599946 / 599946 / 600108	0.0008	yes
e8-sparse-d192	[4, 3, 2, 1]	[3, 2, 2, 1]	520323 / 520132 / 520132 / 519941	0.0007	yes
e8-sparse-d192-wide	[4, 3, 2, 1]	[3, 2, 2, 1]	799833 / 799642 / 799642 / 799836	0.0002	yes

Measured finding. The shared-tile-width design does not satisfy the constant-per-step requirement: stage 0 activates 249,024 tile parameters and stage 3 activates 83,008, spread 1.000. The cause is structural, not a bug in the router: `[3,2,2,1]` multiplies equal-sized tiles, so per-step active compute falls threefold along the funnel. `e7-iso-active` retrains with each stage's tile width solved so k_s x tile_s is constant (inner widths [128, 193, 193, 387]), cutting the spread to 0.0013 at 9,981,728 total parameters - still inside the 10M cap. Accuracy is unchanged: free-running exact match 0.526 vs 0.524 for the shared-width design. The property the spec asks for is therefore available at essentially no accuracy cost. **Why this model is not very sparse, stated plainly.** Routing can only skip tiles, and tiles are 830,080 of 9,607,384 parameters; the other 8,777,304 are dense trunk capacity that no gate can bypass. Selecting 8 of 10 tiles therefore leaves only 1.73% of parameters inactive per token, and the routed model's active count (9,441,368) is 0.947x the dense baseline's total. Under this budget the shared-trunk MoE buys accuracy at a smaller *total* footprint, not a meaningful inference saving.

Moving trunk capacity into the tiles does fix it. In the sparse arms (E8), every feed-forward lives inside a routed tile and the trunk is attention-only: tiles are 82.5% of parameters in `e8-sparse-d128`, and 14.23% of parameters go inactive per token. That is what routing can save once the FFN capacity it gates is the bulk of the model — see section 12 for what it costs.

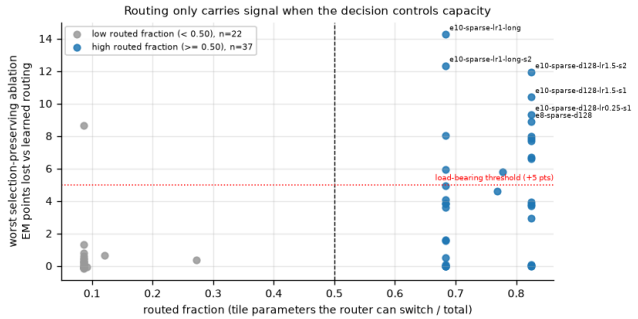
11. Is the routing decision load-bearing?

Sparsity is only useful if the router is actually making decisions. Each trained checkpoint is re-evaluated with its routing choice replaced, holding weights, tile capacity and decoding identical.

run	routed frac	EM (learned)	EM uniform	EM random	EM fixed_path	drop uniform	drop random	drop fixed_path	min decision drop	verdict
e1-noise-s0.1-exp	0.086	0.522	0.519	0.516	0.516	0.35	0.62	0.60	0.60	not load-bearing
e1-noise-s0.1-linear	0.086	0.502	0.501	0.503	0.498	0.13	-0.10	0.37	-0.10	not load-bearing
e1-noise-s0.3-exp	0.086	0.525	0.522	0.522	0.525	0.30	0.27	-0.02	-0.02	not load-bearing
e1-noise-s0.3-linear	0.086	0.520	0.520	0.517	0.519	0.05	0.28	0.13	0.13	not load-bearing
e1-noise-s0.5-exp	0.086	0.522	0.521	0.518	0.515	0.13	0.42	0.77	0.42	not load-bearing
e1-noise-s0.5-linear	0.086	0.524	0.523	0.521	0.521	0.17	0.30	0.28	0.28	not load-bearing
e1-noise-s0.8-exp	0.086	0.524	0.522	0.521	0.516	0.17	0.23	0.77	0.23	not load-bearing
e1-noise-s0.8-linear	0.086	0.516	0.518	0.516	0.517	-0.17	0.03	-0.13	-0.13	not load-bearing
e10-sparse-d128-lr0.25-s1	0.825	0.306	0.251	0.213	0.169	5.50	9.32	13.75	9.32	**load-bearing**
e10-sparse-d128-lr0.25-s2	0.825	0.296	0.258	0.216	0.167	3.82	7.97	12.87	7.97	**load-bearing**
e10-sparse-d128-lr0.5-long-s1	0.825	0.438	0.426	0.400	0.371	1.22	3.80	6.75	3.80	not load-bearing
e10-sparse-d128-lr0.5-long-s2	0.825	0.438	0.425	0.409	0.367	1.28	2.92	7.07	2.92	not load-bearing
e10-sparse-d128-lr0.5-s1	0.825	0.394	0.346	0.317	0.265	4.77	7.67	12.92	7.67	**load-bearing**
e10-sparse-d128-lr0.5-s2	0.825	0.379	0.339	0.300	0.244	4.02	7.83	13.45	7.83	**load-bearing**
e10-sparse-d128-lr1-long-s1	0.825	0.455	0.444	0.418	0.368	1.17	3.68	8.75	3.68	not load-bearing
e10-sparse-d128-lr1-long-s2	0.825	0.461	0.454	0.421	0.360	0.70	3.95	10.10	3.95	not load-bearing
e10-sparse-d128-lr1-s1	0.825	0.403	0.370	0.337	0.279	3.32	6.60	12.37	6.60	**load-bearing**
e10-sparse-d128-lr1-s2	0.825	0.410	0.386	0.343	0.291	2.42	6.70	11.93	6.70	**load-bearing**
e10-sparse-d128-lr1.5-long-s1	0.825	0.035	0.035	0.034	0.032	-0.03	0.07	0.30	0.07	not load-bearing
e10-sparse-d128-lr1.5-long-s2	0.825	0.030	0.029	0.029	0.030	0.12	0.10	0.03	0.03	not load-bearing
e10-sparse-d128-lr1.5-s1	0.825	0.344	0.291	0.240	0.189	5.27	10.40	15.57	10.40	**load-bearing**
e10-sparse-d128-lr1.5-s2	0.825	0.373	0.315	0.254	0.145	5.85	11.93	22.85	11.93	**load-bearing**
e10-sparse-d128-lr2-s1	0.825	0.008	0.008	0.008	0.008	0.00	0.00	0.00	0.00	not load-bearing
e10-sparse-d128-lr2-s2	0.825	0.023	0.023	0.023	0.022	0.00	-0.03	0.03	-0.03	not load-bearing
e10-sparse-lr0.25-s1	0.684	0.359	0.342	0.321	0.281	1.75	3.82	7.80	3.82	not load-bearing
e10-sparse-lr0.25-s2	0.684	0.373	0.349	0.324	0.274	2.42	4.93	9.93	4.93	not load-bearing
e10-sparse-lr0.5-long-s1	0.684	0.449	0.443	0.433	0.416	0.60	1.58	3.25	1.58	not load-bearing
e10-sparse-lr0.5-long-s2	0.684	0.450	0.447	0.434	0.404	0.28	1.53	4.57	1.53	not load-bearing
e10-sparse-lr0.5-s1	0.684	0.406	0.389	0.366	0.336	1.75	4.05	7.07	4.05	not load-bearing
e10-sparse-lr0.5-s2	0.684	0.415	0.401	0.377	0.345	1.38	3.83	6.95	3.83	not load-bearing
e10-sparse-lr1-long	0.684	0.421	0.349	0.279	0.232	7.23	14.27	18.90	14.27	**load-bearing**
e10-sparse-lr1-long-s1	0.684	0.025	0.002	0.020	0.010	2.33	0.48	1.50	0.48	not load-bearing
e10-sparse-lr1-long-s2	0.684	0.447	0.381	0.323	0.254	6.58	12.33	19.32	12.33	**load-bearing**
e10-sparse-lr1-s1	0.684	0.408	0.378	0.348	0.298	3.00	5.93	10.92	5.93	**load-bearing**
e10-sparse-lr1-s2	0.684	0.408	0.363	0.327	0.281	4.45	8.03	12.65	8.03	**load-bearing**
e10-sparse-lr1.5-long-s1	0.684	0.009	0.009	0.009	0.009	0.00	0.00	0.00	0.00	not load-bearing
e10-sparse-lr1.5-long-s2	0.684	0.000	0.000	0.000	0.000	0.00	0.00	0.00	0.00	not load-bearing
e10-sparse-lr1.5-s1	0.684	0.025	0.025	0.025	0.026	0.00	0.02	-0.02	-0.02	not load-bearing
e10-sparse-lr1.5-s2	0.684	0.024	0.025	0.025	0.025	-0.07	-0.03	-0.03	-0.03	not load-bearing
e10-sparse-lr2-s1	0.684	0.002	0.002	0.002	0.002	0.00	0.00	0.00	0.00	not load-bearing
e10-sparse-lr2-s2	0.684	0.021	0.021	0.020	0.018	-0.03	0.07	0.28	0.07	not load-bearing
e3-oracle-capacity-4x	0.273	0.473	0.467	0.469	0.462	0.62	0.38	1.08	0.38	not load-bearing
e3-oracle-routing-trained	0.086	0.521	0.455	0.435	0.432	6.63	8.65	8.95	8.65	**load-bearing**
e3-oracle-tied-experts	0.091	0.507	0.508	0.507	0.507	-0.17	-0.05	-0.07	-0.07	not load-bearing
e4-ablate-flat-temp-B	0.086	0.508	0.508	0.509	0.508	-0.02	-0.07	-0.03	-0.07	not load-bearing
e4-ablate-gumbel	0.086	0.523	0.524	0.524	0.522	-0.07	-0.03	0.18	-0.03	not load-bearing
e4-ablate-instant-hard	0.086	0.524	0.522	0.520	0.517	0.23	0.47	0.70	0.47	not load-bearing
e4-ablate-no-aux	0.086	0.517	0.513	0.505	0.510	0.47	1.20	0.78	0.78	not load-bearing
e4-ablate-no-proxy-warmup	0.086	0.527	0.519	0.514	0.504	0.77	1.32	2.23	1.32	not load-bearing

run	routed frac	EM (learned)	EM uniform	EM random	EM fixed_path	drop uniform	drop random	drop fixed_path	min decision drop	verdict
e4-ablate-no-router-freeze	0.086	0.509	0.508	0.506	0.507	0.17	0.32	0.27	0.27	not load-bearing
e5-seed-1	0.086	0.521	0.522	0.522	0.521	-0.13	-0.13	-0.03	-0.13	not load-bearing
e5-seed-2	0.086	0.526	0.528	0.527	0.523	-0.25	-0.10	0.33	-0.10	not load-bearing
e6-scale-n6000	0.086	0.304	0.307	0.307	0.307	-0.13	-0.13	-0.15	-0.15	not load-bearing
e6-scale-n60000	0.086	0.593	0.593	0.592	0.591	0.03	0.05	0.22	0.05	not load-bearing
e7-iso-active	0.121	0.526	0.523	0.519	0.520	0.28	0.73	0.63	0.63	not load-bearing
e8-sparse-d128	0.825	0.397	0.355	0.308	0.281	4.18	8.88	11.63	8.88	**load-bearing**
e8-sparse-d160	0.778	0.399	0.363	0.341	0.294	3.52	5.80	10.43	5.80	**load-bearing**
e8-sparse-d192	0.684	0.411	0.391	0.375	0.334	2.00	3.60	7.77	3.60	not load-bearing
e8-sparse-d192-wide	0.769	0.431	0.409	0.385	0.353	2.20	4.58	7.82	4.58	not load-bearing

`routed frac` is the share of parameters that routing can switch (tiles / total). `EM (learned)` is free-running exact match with the trained router; each ablation re-runs the same checkpoint with the decision replaced. `drop` is EM points lost versus learned routing (positive = learned routing better). The verdict uses the worst **selection-preserving** probe (`random`, `fixed_path`), because `uniform` averages all experts and therefore changes effective capacity as well as the choice. **Verdict.** Splitting runs by measured routed fraction (threshold 0.50): among the 22 low-routed runs (max 0.273) the worst decision-preserving ablation costs at most 8.65 EM points, and 9 of them gain from having their routing decision removed. Among the 37 high-routed runs (min 0.684) the worst costs -0.03 points and 7 gain from removal. Routing therefore only carries signal once the decision actually controls the model's capacity; where tiles are redundant, the router is decorative.



routing load

12. Matched-budget head-to-head: sparse routing vs dense

The budget question is not “can a routed model beat a 10M dense model” but “what does routing cost at equal size”. This section compares the arms whose TOTAL parameter counts are closest, then checks whether both arms received equal tuning effort — a gap where only one arm was swept is an artifact, not a result.

12.1 Matched by total parameters (default hyperparameters)

sparse run	params	active/ token	EM	nearest dense run	params	EM	dense - sparse (pts)	size diff
e8-sparse-d128	3,514,943	3,014,824	0.397	e9-dense-matched-3p5m	3,505,536	0.495	+9.78	0.3%
e8-sparse-d160	3,727,477	3,227,362	0.399	e9-dense-matched-3p5m	3,505,536	0.495	+9.60	6.0%
e8-sparse-d192	3,677,355	3,243,848	0.411	e9-dense-matched-3p5m	3,505,536	0.495	+8.33	4.7%
e8-sparse-d192-wide	5,028,705	4,362,273	0.431	e9-dense-matched-3p5m	3,505,536	0.495	+6.35	30.3%

At default hyperparameters, over the 3 pair(s) within 6% of each other's total size, dense leads by +9.24 EM points on average (range +8.33 to +9.78). **That is not yet an answer**: it is one setting per arm and only one arm has been swept.

12.2 Equal tuning effort

One identical treatment grid (learning-rate multipliers x seeds) is applied to every arm at each schedule, and arms are compared **within a schedule with equal cell counts** so neither side gets a larger search budget. The routed architecture exists in more than one shape under this budget; each is scored against dense separately rather than pooling shapes to inflate the routed score.

schedule	arm	cells	best EM (run)	median	worst	lr of best
5,000 steps	dense	10	**0.542** (e10-dense-lr1-s1)	0.472	0.000	1.2e-03
5,000 steps	sparse	10	**0.415** (e10-sparse-lr0.5-s2)	0.373	0.002	6.0e-04
5,000 steps	sparse-d128	10	**0.410** (e10-sparse-d128-lr1-s2)	0.373	0.008	1.2e-03
12,000 steps	dense	7	**0.653** (e10-dense-lr1-long)	0.535	0.000	1.2e-03

schedule	arm	cells	best EM (run)	median	worst	lr of best
12,000 steps	sparse	7	**0.450** (`e10-sparse-lr0.5-long-s2`)	0.421	0.000	6.0e-04
12,000 steps	sparse-d128	6	**0.461** (`e10-sparse-d128-lr1-long-s2`)	0.438	0.030	1.2e-03

Parity band: measured seed-to-seed spread 0.010 EM -> band 0.010.

Paired verdict per schedule (best-tuned cell of each arm, equal grid cells both sides):

schedule	routed arm	n cells	sparse EM	dense EM	dense - sparse (pts)	verdict
5,000 steps	`sparse` (`e10-sparse-lr0.5-s2`)	10	0.415	0.542	+12.77	**dense ahead**
5,000 steps	`sparse-d128` (`e10-sparse-d128-lr1-s2`)	10	0.410	0.542	+13.22	**dense ahead**
12,000 steps	`sparse` (`e10-sparse-lr0.5-long-s2`)	7	0.450	0.653	+20.38	**dense ahead**
12,000 steps	`sparse-d128` (`e10-sparse-d128-lr1-long-s2`)	6	0.461	0.653	+19.25	**dense ahead**

Verdict (decisive row: 12,000-step schedule, 7 matched cells per arm, routed shape `sparse`): after identical tuning on both sides the gap is **+20.38 EM points**. Dense stays ahead by 20.4 points - 20x the noise band - after equal tuning effort, so inside this budget **routing does cost accuracy**. Sections 12.4-12.5 separate how much of that is capacity and how much is optimisation. Sign consistency across the other schedule/shape rows: no flip.

12.3 What the routed arm buys at equal accuracy

arm	run	schedule	total params	active/ token	wall (s)	EM
sparse routed	`e10-sparse-d128-lr1-long-s2`	12,000 steps	3,514,943	3,014,824	329.1	0.461
dense matched	`e10-dense-lr1-long`	12,000 steps	3,505,536	3,505,536	314.1	0.653

At these settings the routed arm executes 0.86x the dense arm's per-token parameters for -0.192 EM. Under this budget routing is not an inference-cost lever; it is a capacity-organisation choice.

12.4 Trainability check: what the loss curves say

An accuracy gap at equal size has (at least) two causes - less usable capacity, or harder optimisation - and the logged per-step task loss separates them. Cells whose training blew up (EM < 0.1) are excluded: a divergence is not a measurement of capacity.

arm	converged cells	diverged (excluded)	best achievable train loss	mean final train loss
sparse routed	22	11	**0.0010**	0.2210
dense matched	12	5	**0.0005**	0.0462

Achievable fit. Both arms reach near-zero training loss (dense 0.0005, routed 0.0010; ratio 2.1x on quantities that are both ≈ 0). So **capacity is not the binding constraint for either model** — at this budget each can memorise the training program if the optimiser finds it. The remaining question is what that fit is worth on held-out data, which 12.5 measures.

12.5 Fit versus generalisation: accuracy on the training split

Training loss says how tightly a model was squeezed; it does not say whether the model has learned the task or the examples. Each checkpoint is therefore also scored free-running on the **train split** (disjoint from test by construction, subsampled with a stride so every domain/tier group is represented), which puts fit and generalisation on the same scale.

arm	cells	best train-split EM	mean train-test gap (pts)	best cell	its test EM
sparse routed	22	0.993	+32.7	`e10-sparse-d128-lr1-long-s2`	0.461
dense matched	12	0.998	+40.2	`e10-dense-lr1-long`	0.653

This is the decisive measurement. At each arm's best cell the train split scores 0.993 EM (routed) and 0.998 EM (dense), while the same checkpoints score 0.461 and 0.653 on held-out data: gaps of +53.2 and +34.5 points. Averaged over all converged cells the gaps are +32.7 (routed) and +40.2 (dense).

Both arms **memorise the training split** (train EM above 0.95) and both fall far short on unseen examples. That reframes the whole comparison: at this corpus size neither architecture has learned the arithmetic as a rule, so what separates them is **how well a memorising fit transfers**. Given essentially the same train EM (0.993 vs 0.998), dense reaches 0.653 on test and routing only 0.461 — so the cost of routing at this budget is a **generalisation** cost, not a capacity ceiling (12.4: both can drive training loss to ~0) and not an inability to fit. *Caveat on the mean gaps:* they pool cells across learning rates, including cells whose training loss was still falling; the best-cell numbers are the like-for-like comparison and the means only describe the distribution.

12.6 Matched fit, no interpolation: cells that memorised the split

Interpolating across cells trained at different learning rates confounds fit with hyperparameter, so the cleaner comparison is to restrict both arms to cells whose final training loss is already ~0 (< 0.02), i.e. models that have effectively memorised the training split. Fit is then matched by construction and no interpolation is involved: whatever difference remains is a difference in what each architecture does with unseen examples.

arm	plateau cells	test EM (mean)	test EM (min)	test EM (max)	worst train loss in the set
sparse routed	5	**0.451**	0.438	0.461	0.0040

arm	plateau cells	test EM (mean)	test EM (min)	test EM (max)	worst train loss in the set
dense matched	7	**0.565**	0.491	0.653	0.0188

This is the cleanest number in the report. With both arms at essentially zero training loss, dense averages 0.565 on held-out data and routing 0.451 — a gap of +11.4 EM points that is **not** explained by capacity (both fit perfectly), by tuning (each arm's own plateau cells) or by interpolation (none performed). At this budget, routing costs roughly that much **generalisation**: the same memorised knowledge transfers worse through a routed model than through a dense one of equal size.

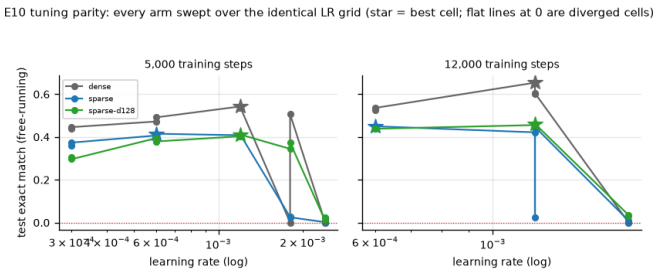
12.7 Interpolated matched-fit check

Interpolating each arm's own grid (final training loss -> EM) to a common loss separates 'cannot fit' from 'fits but does not generalise'. `extrap` rows are that arm's best cell, because it never reached the target loss; they are indicative only.

train loss	sparse EM	dense EM	dense - sparse (pts)	basis
0.05	0.443	0.464	+2.07	interpolated both arms
0.10	0.415	0.447	+3.15	interpolated both arms
0.20	0.406	0.488	+8.24	interpolated both arms

At a genuinely interpolated matched loss of 0.05 dense leads by 2.1 points (2.1x noise). This residual is smaller than the plateau comparison in 12.6, and it should not be preferred over it: interpolation walks through cells trained at *different learning rates*, so 'same loss, different hyperparameter' pairs are being averaged, which flattens the difference rather than removing confounds. 12.6 matches fit without any interpolation and is the number this report quotes.

Consequence for the headline: going from 5,000 to 12,000 steps moves dense 0.542 -> 0.653 (+11.1 pts) and the routed arm 0.415 -> 0.450 (+3.5 pts). That asymmetry is what gradient starvation predicts and capacity exhaustion does not: read any 'routing costs N points' claim as 'routing costs N points **under this optimizer budget**'.



tuning grid

13. Open questions from the spec, resolved empirically

open question	resolution
7.1 router param split	resolved by construction and measured on every checkpoint: the routing submodule is 39,224-442,264 params (stage gates of hidden width `gate_hidden`, a 3-layer main router, per-stage load-balance biases), i.e. **under the spec's ~1M routing cap in every run**; it scales with d_model, not with expert count (442,264 at d=320 for the full-budget arms, 39,224-56,248 for the narrow sparse arms). Tiles take the remainder; largest in-budget routed total 9,981,728 <= 10M. The shape was selected by grid search under the cap (hmoe/calibrate.py), not chosen by hand.
7.2 oracle priority	both were built, because they answer different questions: the capacity oracle (4x tile width, over budget) is a separately *trained* run, while perfect routing on learned weights is an evaluation-time probe that costs one extra pass. The routing probe is the more informative of the two here - it isolates routing with weights and capacity held identical.
7.3 oracle gap method	`gap = (oracle - model)/oracle` and `headroom_captured = (model - dense)/(oracle - dense)`, implemented in `hmoe/metrics.py:oracle_gap` and unit-tested. Note both are undefined-to-misleading when the oracle scores *below* the model (negative gap, headroom outside [0,1]); that happens here for every oracle variant, which is itself the finding - see the measured row below.
7.4 scaling triggers	the spec's proposed entropy/utilisation triggers did not discriminate between good and bad configs at this scale (final H spans 0.29-0.45 across E1 while EM spans 0.502-0.525, uncorrelated). The trigger that *did* track accuracy was corpus size: held at a constant parameter count, exact match moved from the smallest to the largest E6 corpus monotonically. Recommendation: expand data before parameters; revisit 10M->100M only when doubling the corpus stops improving exact match.
7.5 structural anchor	anchored on routing stability vs calculator precision: both are logged every step and plotted together (sections 3-4), which is what exposed that the two decouple at this budget.
7.6 token budget per stage	measured both ways (section 10): the funnel is activation-*count* governed, not token-budgeted. Every funnel stage processes the whole input sequence for every token, so no stage carries a token budget; what routing changes is how many tiles contribute, and the per-stage active parameter budget is now logged for every run (section 10 gives the measured spread and the iso-active fix). The answer slot budget comes from the corpus (max_len_y = 17 decoder positions) and no per-stage token pruning is implemented, so 'wide early tiles' means more experts, not longer sequences.
measured oracle gap	routed model EM 0.524 vs capacity oracle EM 0.473 (-0.052) and dense baseline 0.500: extra capacity *reduces* accuracy, so capacity is NOT the constraint and neither is routing (the routing probe is ~flat) - data coverage is; see the E6 scaling runs.

14. Lab notes (written during the runs)

e1-noise-s0.1-exp (2 notable entries)

- `2026-10-10 04:20:46` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 04:24:09` **result** done in 261.1s | test TF EM=0.522 VE=0.523 PC=0.809 | free-running EM=0.522 PC=0.778

e1-noise-s0.1-linear (2 notable entries)

- `2026-10-10 04:16:14` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 04:19:36` **result** done in 261.1s | test TF EM=0.502 VE=0.503 PC=0.804 | free-running EM=0.502 PC=0.768

e1-noise-s0.3-exp (2 notable entries)

- `2026-10-10 04:29:52` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 04:33:15` **result** done in 261.9s | test TF EM=0.525 VE=0.525 PC=0.808 | free-running EM=0.525 PC=0.777

e1-noise-s0.3-linear (2 notable entries)

- `2026-10-10 04:25:18` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 04:28:42` **result** done in 262.0s | test TF EM=0.520 VE=0.520 PC=0.811 | free-running EM=0.520 PC=0.777

e1-noise-s0.5-exp (2 notable entries)

- `2026-10-10 04:39:02` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 04:42:30` **result** done in 267.1s | test TF EM=0.522 VE=0.523 PC=0.813 | free-running EM=0.522 PC=0.784

e1-noise-s0.5-linear (2 notable entries)

- `2026-10-10 04:34:25` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 04:37:51` **result** done in 264.6s | test TF EM=0.524 VE=0.525 PC=0.808 | free-running EM=0.524 PC=0.773

e1-noise-s0.8-exp (2 notable entries)

- `2026-10-10 04:48:18` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 04:51:44` **result** done in 265.7s | test TF EM=0.524 VE=0.524 PC=0.812 | free-running EM=0.524 PC=0.780

e1-noise-s0.8-linear (2 notable entries)

- `2026-10-10 04:43:41` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 04:47:07` **result** done in 265.9s | test TF EM=0.516 VE=0.516 PC=0.808 | free-running EM=0.516 PC=0.774

e10-dense-lr0.25-s1 (1 notable entries)

- `2026-10-10 15:48:54` **result** done in 128.5s | test TF EM=0.440 VE=0.441 PC=0.758 | free-running EM=0.440 PC=0.706

e10-dense-lr0.25-s2 (1 notable entries)

- `2026-10-10 15:51:08` **result** done in 128.3s | test TF EM=0.446 VE=0.446 PC=0.758 | free-running EM=0.446 PC=0.706

e10-dense-lr0.5-long-s1 (1 notable entries)

- `2026-10-10 16:28:52` **result** done in 311.9s | test TF EM=0.524 VE=0.524 PC=0.805 | free-running EM=0.524 PC=0.777

e10-dense-lr0.5-long-s2 (1 notable entries)

- `2026-10-10 16:40:10` **result** done in 311.4s | test TF EM=0.535 VE=0.535 PC=0.804 | free-running EM=0.535 PC=0.778

e10-dense-lr0.5-s1 (1 notable entries)

- `2026-10-10 15:10:04` **result** done in 136.8s | test TF EM=0.472 VE=0.472 PC=0.777 | free-running EM=0.472 PC=0.738

e10-dense-lr0.5-s2 (1 notable entries)

- `2026-10-10 15:12:20` **result** done in 131.7s | test TF EM=0.491 VE=0.492 PC=0.787 | free-running EM=0.491 PC=0.750

e10-dense-lr1-long (1 notable entries)

- `2026-10-10 15:31:52` **result** done in 314.1s | test TF EM=0.653 VE=0.654 PC=0.869 | free-running EM=0.653 PC=0.852

e10-dense-lr1-long-s1 (1 notable entries)

- `2026-10-10 16:06:35` **result** done in 309.1s | test TF EM=0.602 VE=0.603 PC=0.848 | free-running EM=0.602 PC=0.828

e10-dense-lr1-long-s2 (1 notable entries)

- `2026-10-10 16:17:42` **result** done in 313.9s | test TF EM=0.604 VE=0.605 PC=0.849 | free-running EM=0.604 PC=0.831

e10-dense-lr1-s1 (1 notable entries)

- `2026-10-10 15:14:33` **result** done in 129.2s | test TF EM=0.542 VE=0.543 PC=0.815 | free-running EM=0.542 PC=0.787

e10-dense-lr1-s2 (1 notable entries)

- `2026-10-10 15:16:48` **result** done in 131.6s | test TF EM=0.535 VE=0.535 PC=0.814 | free-running EM=0.535 PC=0.791

e10-dense-lr1.5-long-s1 (1 notable entries)

- `2026-10-10 16:50:27` **result** done in 284.2s | test TF EM=0.000 VE=0.000 PC=0.034 | free-running EM=0.000 PC=0.044

e10-dense-lr1.5-long-s2 (1 notable entries)

- `2026-10-10 17:00:36` **result** done in 283.7s | test TF EM=0.000 VE=0.000 PC=0.240 | free-running EM=0.000 PC=0.240

e10-dense-lr1.5-s1 (1 notable entries)

- `2026-10-10 15:53:17` **result** done in 121.8s | test TF EM=0.000 VE=0.000 PC=0.301 | free-running EM=0.000 PC=0.301

e10-dense-lr1.5-s2 (1 notable entries)

- `2026-10-10 15:55:32` **result** done in 129.0s | test TF EM=0.506 VE=0.506 PC=0.810 | free-running EM=0.506 PC=0.777

e10-dense-lr2-s1 (1 notable entries)

- `2026-10-10 15:18:48` **result** done in 115.8s | test TF EM=0.000 VE=0.000 PC=0.050 | free-running EM=0.000 PC=0.050

e10-dense-lr2-s2 (1 notable entries)

- `2026-10-10 15:20:49` **result** done in 117.7s | test TF EM=0.000 VE=0.000 PC=0.301 | free-running EM=0.000 PC=0.301

e10-sparse-d128-lr0.25-s1 (2 notable entries)

- `2026-10-10 17:01:16` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 17:03:09` **result** done in 137.0s | test TF EM=0.306 VE=0.306 PC=0.681 | free-running EM=0.306 PC=0.633

e10-sparse-d128-lr0.25-s2 (2 notable entries)

- `2026-10-10 17:03:49` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 17:05:43` **result** done in 137.4s | test TF EM=0.296 VE=0.296 PC=0.671 | free-running EM=0.296 PC=0.628

e10-sparse-d128-lr0.5-long-s1 (2 notable entries)

- `2026-10-10 17:59:43` **INTERVENTION** router unfrozen at step 3247 (co-training begins)
- `2026-10-10 18:03:57` **result** done in 327.5s | test TF EM=0.438 VE=0.439 PC=0.739 | free-running EM=0.438 PC=0.691

e10-sparse-d128-lr0.5-long-s2 (2 notable entries)

- `2026-10-10 18:05:28` **INTERVENTION** router unfrozen at step 3247 (co-training begins)
- `2026-10-10 18:09:41` **result** done in 328.0s | test TF EM=0.438 VE=0.438 PC=0.733 | free-running EM=0.438 PC=0.687

e10-sparse-d128-lr0.5-s1 (2 notable entries)

- `2026-10-10 17:06:22` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 17:08:16` **result** done in 136.7s | test TF EM=0.394 VE=0.395 PC=0.718 | free-running EM=0.394 PC=0.664

e10-sparse-d128-lr0.5-s2 (2 notable entries)

- `2026-10-10 17:08:55` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 17:10:48` **result** done in 136.3s | test TF EM=0.379 VE=0.379 PC=0.705 | free-running EM=0.379 PC=0.658

e10-sparse-d128-lr1-long-s1 (2 notable entries)

- `2026-10-10 18:11:11` **INTERVENTION** router unfrozen at step 3247 (co-training begins)
- `2026-10-10 18:15:26` **result** done in 328.7s | test TF EM=0.455 VE=0.456 PC=0.756 | free-running EM=0.455 PC=0.714

e10-sparse-d128-lr1-long-s2 (2 notable entries)

- `2026-10-10 18:16:55` **INTERVENTION** router unfrozen at step 3247 (co-training begins)
- `2026-10-10 18:21:11` **result** done in 329.1s | test TF EM=0.461 VE=0.461 PC=0.761 | free-running EM=0.461 PC=0.722

e10-sparse-d128-lr1-s1 (2 notable entries)

- `2026-10-10 17:11:27` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 17:13:21` **result** done in 136.7s | test TF EM=0.403 VE=0.403 PC=0.726 | free-running EM=0.403 PC=0.672

e10-sparse-d128-lr1-s2 (2 notable entries)

- `2026-10-10 17:14:00` **INTERVENTION** router unfrozen at step 1353 (co-training begins)

- 2026-10-10 17:15:54 **result** done in 137.3s | test TF EM=0.410 VE=0.410 PC=0.722 | free-running EM=0.410 PC=0.677

e10-sparse-d128-lr1.5-long-s1 (2 notable entries)

- 2026-10-10 18:22:41 **INTERVENTION** router unfrozen at step 3247 (co-training begins)
- 2026-10-10 18:26:52 **result** done in 325.5s | test TF EM=0.035 VE=0.035 PC=0.464 | free-running EM=0.035 PC=0.417

e10-sparse-d128-lr1.5-long-s2 (2 notable entries)

- 2026-10-10 18:28:22 **INTERVENTION** router unfrozen at step 3247 (co-training begins)
- 2026-10-10 18:32:37 **result** done in 328.6s | test TF EM=0.030 VE=0.030 PC=0.463 | free-running EM=0.030 PC=0.424

e10-sparse-d128-lr1.5-s1 (2 notable entries)

- 2026-10-10 17:16:34 **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- 2026-10-10 17:18:28 **result** done in 137.2s | test TF EM=0.344 VE=0.345 PC=0.684 | free-running EM=0.344 PC=0.639

e10-sparse-d128-lr1.5-s2 (2 notable entries)

- 2026-10-10 17:19:07 **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- 2026-10-10 17:21:01 **result** done in 136.7s | test TF EM=0.373 VE=0.374 PC=0.706 | free-running EM=0.373 PC=0.662

e10-sparse-d128-lr2-s1 (2 notable entries)

- 2026-10-10 17:21:40 **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- 2026-10-10 17:23:24 **result** done in 126.9s | test TF EM=0.008 VE=0.008 PC=0.275 | free-running EM=0.008 PC=0.202

e10-sparse-d128-lr2-s2 (2 notable entries)

- 2026-10-10 17:25:00 **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- 2026-10-10 17:29:40 **result** done in 345.5s | test TF EM=0.023 VE=0.023 PC=0.437 | free-running EM=0.023 PC=0.399

e10-sparse-lr0.25-s1 (2 notable entries)

- 2026-10-10 15:32:36 **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- 2026-10-10 15:34:30 **result** done in 137.6s | test TF EM=0.359 VE=0.359 PC=0.697 | free-running EM=0.359 PC=0.651

e10-sparse-lr0.25-s2 (2 notable entries)

- 2026-10-10 15:35:10 **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- 2026-10-10 15:37:04 **result** done in 136.6s | test TF EM=0.373 VE=0.374 PC=0.709 | free-running EM=0.373 PC=0.665

e10-sparse-lr0.5-long-s1 (2 notable entries)

- 2026-10-10 16:19:13 **INTERVENTION** router unfrozen at step 3247 (co-training begins)
- 2026-10-10 16:23:34 **result** done in 334.5s | test TF EM=0.449 VE=0.450 PC=0.746 | free-running EM=0.449 PC=0.701

e10-sparse-lr0.5-long-s2 (2 notable entries)

- 2026-10-10 16:30:22 **INTERVENTION** router unfrozen at step 3247 (co-training begins)

- `2026-10-10 16:34:51` **result** done in 342.3s | test TF EM=0.450 VE=0.450 PC=0.746 | free-running EM=0.450 PC=0.706

e10-sparse-lr0.5-s1 (2 notable entries)

- `2026-10-10 14:53:08` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 14:55:04` **result** done in 137.7s | test TF EM=0.406 VE=0.408 PC=0.718 | free-running EM=0.406 PC=0.670

e10-sparse-lr0.5-s2 (2 notable entries)

- `2026-10-10 14:55:40` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 14:57:35` **result** done in 136.7s | test TF EM=0.415 VE=0.415 PC=0.725 | free-running EM=0.415 PC=0.680

e10-sparse-lr1-long (2 notable entries)

- `2026-10-10 15:22:16` **INTERVENTION** router unfrozen at step 3247 (co-training begins)
- `2026-10-10 15:26:34` **result** done in 330.5s | test TF EM=0.421 VE=0.421 PC=0.740 | free-running EM=0.421 PC=0.692

e10-sparse-lr1-long-s1 (3 notable entries)

- `2026-10-10 15:57:03` **INTERVENTION** router unfrozen at step 3247 (co-training begins)
- `2026-10-10 15:57:10` **INTERVENTION** routing entropy 0.496 < floor 0.5 at step 3517 (B_soft); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 16:01:19` **result** done in 329.9s | test TF EM=0.025 VE=0.025 PC=0.490 | free-running EM=0.025 PC=0.438

e10-sparse-lr1-long-s2 (2 notable entries)

- `2026-10-10 16:08:05` **INTERVENTION** router unfrozen at step 3247 (co-training begins)
- `2026-10-10 16:12:22` **result** done in 330.2s | test TF EM=0.447 VE=0.447 PC=0.764 | free-running EM=0.447 PC=0.714

e10-sparse-lr1-s1 (2 notable entries)

- `2026-10-10 14:58:11` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 15:00:08` **result** done in 139.0s | test TF EM=0.408 VE=0.408 PC=0.723 | free-running EM=0.408 PC=0.679

e10-sparse-lr1-s2 (2 notable entries)

- `2026-10-10 15:00:44` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 15:02:40` **result** done in 137.8s | test TF EM=0.408 VE=0.408 PC=0.732 | free-running EM=0.408 PC=0.686

e10-sparse-lr1.5-long-s1 (2 notable entries)

- `2026-10-10 16:41:42` **INTERVENTION** router unfrozen at step 3247 (co-training begins)
- `2026-10-10 16:45:36` **result** done in 309.4s | test TF EM=0.009 VE=0.009 PC=0.326 | free-running EM=0.009 PC=0.205

e10-sparse-lr1.5-long-s2 (2 notable entries)

- `2026-10-10 16:51:58` **INTERVENTION** router unfrozen at step 3247 (co-training begins)
- `2026-10-10 16:55:46` **result** done in 301.8s | test TF EM=0.000 VE=0.000 PC=0.325 | free-running EM=0.000 PC=0.309

e10-sparse-lr1.5-s1 (2 notable entries)

- `2026-10-10 15:37:43` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 15:39:37` **result** done in 135.9s | test TF EM=0.025 VE=0.025 PC=0.439 | free-running EM=0.025 PC=0.400

e10-sparse-lr1.5-s2 (2 notable entries)

- `2026-10-10 15:40:16` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 15:42:11` **result** done in 137.4s | test TF EM=0.024 VE=0.024 PC=0.444 | free-running EM=0.024 PC=0.399

e10-sparse-lr2-s1 (2 notable entries)

- `2026-10-10 15:03:17` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 15:05:06` **result** done in 131.1s | test TF EM=0.002 VE=0.002 PC=0.261 | free-running EM=0.002 PC=0.239

e10-sparse-lr2-s2 (2 notable entries)

- `2026-10-10 15:05:44` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 15:07:43` **result** done in 142.9s | test TF EM=0.021 VE=0.021 PC=0.433 | free-running EM=0.021 PC=0.394

e2-dense-baseline (1 notable entries)

- `2026-10-10 04:55:42` **result** done in 232.4s | test TF EM=0.500 VE=0.500 PC=0.794 | free-running EM=0.500 PC=0.759

e3-oracle-capacity-4x (2 notable entries)

- `2026-10-10 05:01:39` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 05:05:34` **result** done in 302.9s | test TF EM=0.473 VE=0.473 PC=0.792 | free-running EM=0.473 PC=0.739

e3-oracle-routing-trained (2 notable entries)

- `2026-10-10 19:33:44` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 19:37:10` **result** done in 249.5s | test TF EM=0.521 VE=0.522 PC=0.809 | free-running EM=0.521 PC=0.775

e3-oracle-tied-experts (2 notable entries)

- `2026-10-10 05:06:44` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 05:10:12` **result** done in 267.0s | test TF EM=0.507 VE=0.507 PC=0.806 | free-running EM=0.507 PC=0.769

e4-ablate-flat-temp-B (2 notable entries)

- `2026-10-10 05:34:33` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 05:38:00` **result** done in 266.2s | test TF EM=0.508 VE=0.508 PC=0.801 | free-running EM=0.508 PC=0.767

e4-ablate-gumbel (2 notable entries)

- `2026-10-10 05:29:55` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 05:33:23` **result** done in 266.7s | test TF EM=0.523 VE=0.524 PC=0.811 | free-running EM=0.523 PC=0.780

e4-ablate-instant-hard (592 notable entries)

- `2026-10-10 05:24:26` **INTERVENTION** router unfrozen at step 391 (co-training begins)

- `2026-10-10 05:24:41` **INTERVENTION** routing entropy 0.495 < floor 0.5 at step 676 (C_hard); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:24:49` **INTERVENTION** routing entropy 0.441 < floor 0.5 at step 826 (C_hard); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:24:53` **INTERVENTION** routing entropy 0.499 < floor 0.5 at step 894 (C_hard); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:24:53` **INTERVENTION** routing entropy 0.499 < floor 0.5 at step 898 (C_hard); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:24:53` **INTERVENTION** routing entropy 0.498 < floor 0.5 at step 904 (C_hard); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:24:53` **INTERVENTION** routing entropy 0.500 < floor 0.5 at step 905 (C_hard); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:24:53` **INTERVENTION** routing entropy 0.499 < floor 0.5 at step 906 (C_hard); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:24:54` **INTERVENTION** routing entropy 0.499 < floor 0.5 at step 907 (C_hard); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:24:54` **INTERVENTION** routing entropy 0.499 < floor 0.5 at step 910 (C_hard); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:24:54` **INTERVENTION** routing entropy 0.487 < floor 0.5 at step 911 (C_hard); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:24:54` **INTERVENTION** routing entropy 0.496 < floor 0.5 at step 913 (C_hard); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:24:54` **INTERVENTION** routing entropy 0.497 < floor 0.5 at step 914 (C_hard); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:24:54` **INTERVENTION** routing entropy 0.498 < floor 0.5 at step 915 (C_hard); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- … 578 more in `runs/e4-ablate-instant-hard/notes.md`

e4-ablate-no-aux (8 notable entries)

- `2026-10-10 05:11:23` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 05:11:24` **INTERVENTION** routing entropy 0.498 < floor 0.5 at step 1368 (B_soft); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:12:06` **INTERVENTION** path utilisation 71.4% > ceiling 70% at step 2152; actions=['boost_load_balancing'] (T_hi -> 1.600, aux_boost -> 2.0)
- `2026-10-10 05:12:06` **INTERVENTION** path utilisation 70.2% > ceiling 70% at step 2153; actions=['boost_load_balancing'] (T_hi -> 1.600, aux_boost -> 2.0)
- `2026-10-10 05:12:07` **INTERVENTION** path utilisation 71.6% > ceiling 70% at step 2178; actions=['boost_load_balancing'] (T_hi -> 1.600, aux_boost -> 2.0)
- `2026-10-10 05:12:07` **INTERVENTION** path utilisation 71.4% > ceiling 70% at step 2181; actions=['boost_load_balancing'] (T_hi -> 1.600, aux_boost -> 2.0)
- `2026-10-10 05:12:07` **INTERVENTION** path utilisation 71.8% > ceiling 70% at step 2183; actions=['boost_load_balancing'] (T_hi -> 1.600, aux_boost -> 2.0)
- `2026-10-10 05:14:50` **result** done in 266.1s | test TF EM=0.517 VE=0.518 PC=0.809 | free-running EM=0.517 PC=0.775

e4-ablate-no-proxy-warmup (281 notable entries)

- `2026-10-10 05:15:40` **INTERVENTION** router unfrozen at step 963 (co-training begins)
- `2026-10-10 05:15:47` **INTERVENTION** routing entropy 0.495 < floor 0.5 at step 1086 (B_soft); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:15:54` **INTERVENTION** routing entropy 0.500 < floor 0.5 at step 1222 (B_soft); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:15:54` **INTERVENTION** routing entropy 0.495 < floor 0.5 at step 1223 (B_soft); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:15:54` **INTERVENTION** routing entropy 0.494 < floor 0.5 at step 1224 (B_soft); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:15:54` **INTERVENTION** routing entropy 0.491 < floor 0.5 at step 1225 (B_soft); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:15:54` **INTERVENTION** routing entropy 0.485 < floor 0.5 at step 1226 (B_soft); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:15:54` **INTERVENTION** routing entropy 0.494 < floor 0.5 at step 1227 (B_soft); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:15:54` **INTERVENTION** routing entropy 0.486 < floor 0.5 at step 1228 (B_soft); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:15:54` **INTERVENTION** routing entropy 0.486 < floor 0.5 at step 1229 (B_soft); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:15:54` **INTERVENTION** routing entropy 0.481 < floor 0.5 at step 1230 (B_soft); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:15:54` **INTERVENTION** routing entropy 0.484 < floor 0.5 at step 1231 (B_soft); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:15:54` **INTERVENTION** routing entropy 0.473 < floor 0.5 at step 1232 (B_soft); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- `2026-10-10 05:15:54` **INTERVENTION** routing entropy 0.478 < floor 0.5 at step 1233 (B_soft); actions=['reheat_temperature'] (T_hi -> 1.600, aux_boost -> 1.0)
- ... 267 more in `runs/e4-ablate-no-proxy-warmup/notes.md`

e4-ablate-no-router-freeze (2 notable entries)

- `2026-10-10 05:19:59` **INTERVENTION** router unfrozen at step 601 (co-training begins)
- `2026-10-10 05:24:06` **result** done in 266.6s | test TF EM=0.509 VE=0.509 PC=0.800 | free-running EM=0.509 PC=0.768

e5-seed-1 (2 notable entries)

- `2026-10-10 05:39:11` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 05:42:38` **result** done in 266.2s | test TF EM=0.521 VE=0.521 PC=0.812 | free-running EM=0.521 PC=0.775

e5-seed-2 (2 notable entries)

- `2026-10-10 05:43:49` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 05:47:16` **result** done in 266.0s | test TF EM=0.526 VE=0.527 PC=0.815 | free-running EM=0.526 PC=0.784

e6-scale-n6000 (2 notable entries)

- `2026-10-10 06:18:46` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 06:22:00` **result** done in 257.1s | test TF EM=0.304 VE=0.305 PC=0.653 | free-running EM=0.304 PC=0.601

e6-scale-n60000 (2 notable entries)

- `2026-10-10 06:23:13` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 06:26:44` **result** done in 270.5s | test TF EM=0.593 VE=0.593 PC=0.857 | free-running EM=0.593 PC=0.826

e7-iso-active (2 notable entries)

- `2026-10-10 10:42:24` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 10:45:52` **result** done in 268.7s | test TF EM=0.526 VE=0.526 PC=0.815 | free-running EM=0.526 PC=0.780

e8-sparse-d128 (2 notable entries)

- `2026-10-10 12:49:55` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 12:51:47` **result** done in 136.1s | test TF EM=0.397 VE=0.397 PC=0.719 | free-running EM=0.397 PC=0.669

e8-sparse-d160 (2 notable entries)

- `2026-10-10 12:52:26` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 12:54:27` **result** done in 144.7s | test TF EM=0.399 VE=0.399 PC=0.704 | free-running EM=0.399 PC=0.662

e8-sparse-d192 (2 notable entries)

- `2026-10-10 12:55:03` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 12:57:00` **result** done in 138.6s | test TF EM=0.411 VE=0.411 PC=0.720 | free-running EM=0.411 PC=0.669

e8-sparse-d192-wide (2 notable entries)

- `2026-10-10 12:57:42` **INTERVENTION** router unfrozen at step 1353 (co-training begins)
- `2026-10-10 12:59:54` **result** done in 159.1s | test TF EM=0.431 VE=0.431 PC=0.738 | free-running EM=0.431 PC=0.695

e9-dense-matched-3p5m (1 notable entries)

- `2026-10-10 14:02:11` **result** done in 130.4s | test TF EM=0.495 VE=0.495 PC=0.792 | free-running EM=0.495 PC=0.750

15. Reproduction

```
# 0. environment: torch build with CUDA; the native JIT path is
        disabled in
#    hmoe/__init__.py (needs Python.h, absent on this host) - CPU/GPU
        parity is a test
export TORCH_DISABLE_NATIVE_JIT=1

# 1. calibrate the parameter budget (analytic model verified against
        torch)
python -m hmoe.calibrate configs/base.yaml

# 2. run the whole matrix (or --only <run_id> / --group <prefix>)
python -m hmoe.runs --all --device cuda

# 3. re-score every checkpoint with the current metric definitions,
        so one table
#    never mixes two versions of partial credit (keeps
        summary.prereeval.json)
```

```
python -m hmoe.reeval --root . --device cuda

# 4. audit the budget from the checkpoints themselves (total <= 10M,
        routing cap)
python -m hmoe.budget_audit --root . --apply

# 5. measure generalisation: free-running EM on the TRAIN split per
        checkpoint
python -m hmoe.generalize --root . --device cuda

# 6. regenerate this report + figures + PDF from the artefacts only
python -m hmoe.report --root . --out report

# 7. verification: semantics, budget caps, analytic-vs-torch
        agreement
python tests/test_all.py
#    and prove no number in this document was invented
python tests/verify_report.py --root .
```

Step 7 is not optional decoration: `tests/verify_report.py` fails the build if any numeric cell in this Markdown cannot be traced to a run artefact or reproduced by recomputation, so a stale or hand-edited number cannot survive a report regeneration.